

BFS Traversal

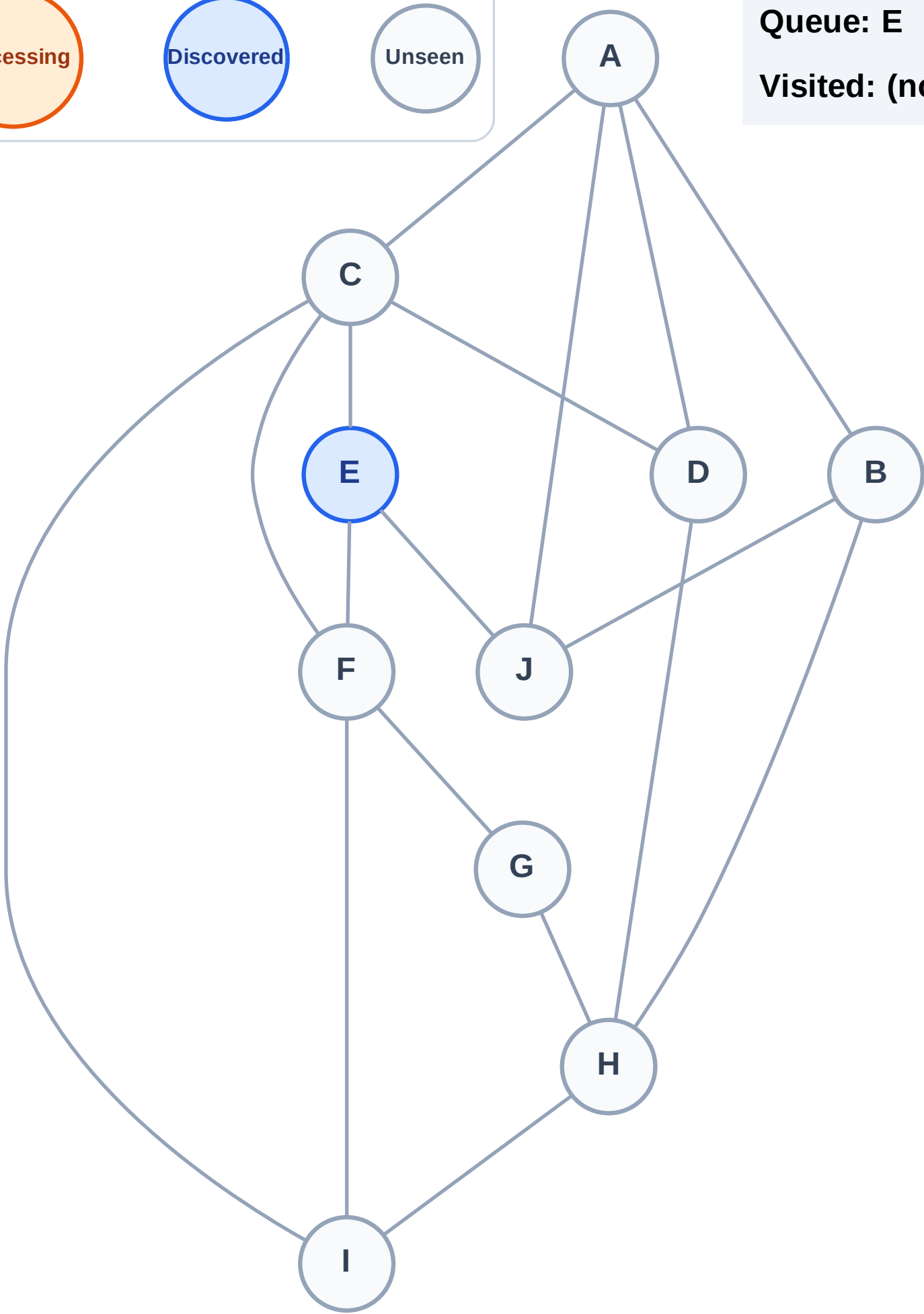
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

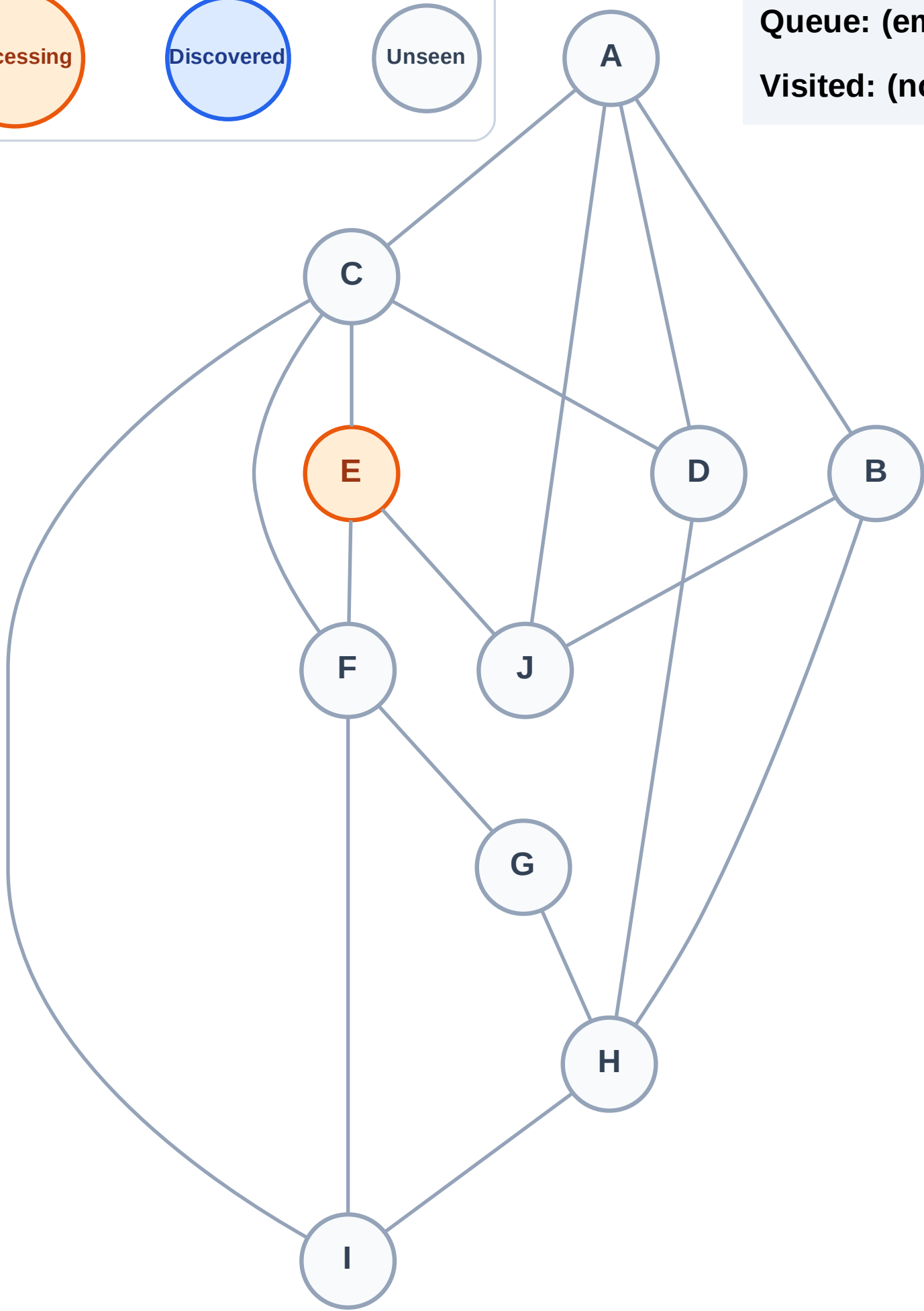
Step 2: Process node E

Legend



Queue: (empty)

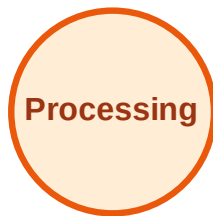
Visited: (none)



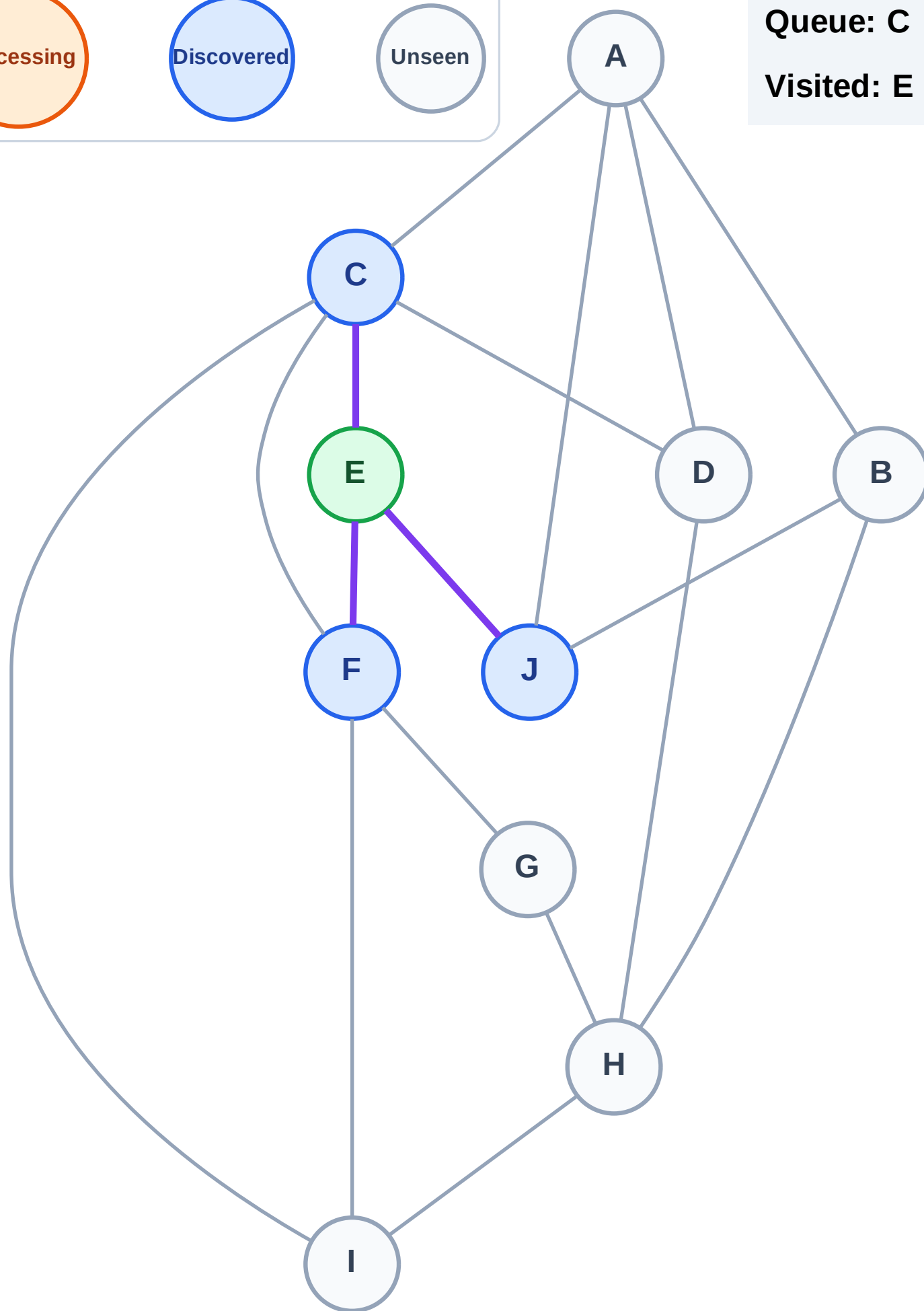
BFS Traversal

Step 3: Discover C, F, J from E

Legend



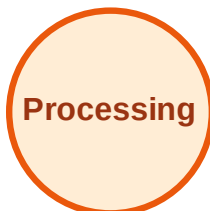
Queue: C → F → J
Visited: E



BFS Traversal

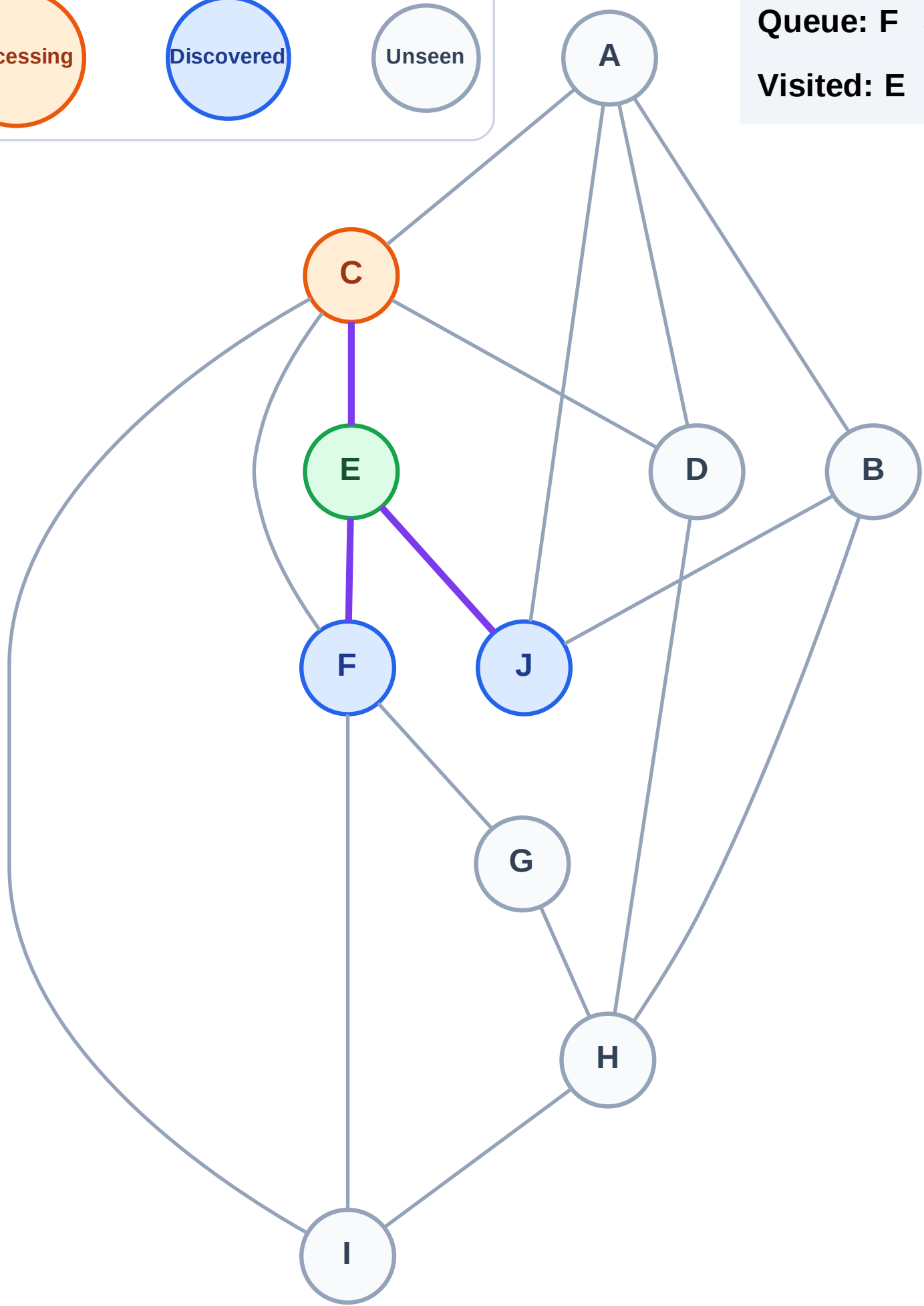
Step 4: Process node C

Legend



Queue: F → J

Visited: E



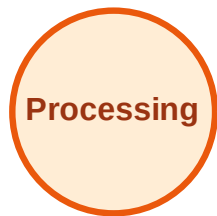
BFS Traversal

Step 5: Discover A, D, I from C

Legend



Complete



Processing



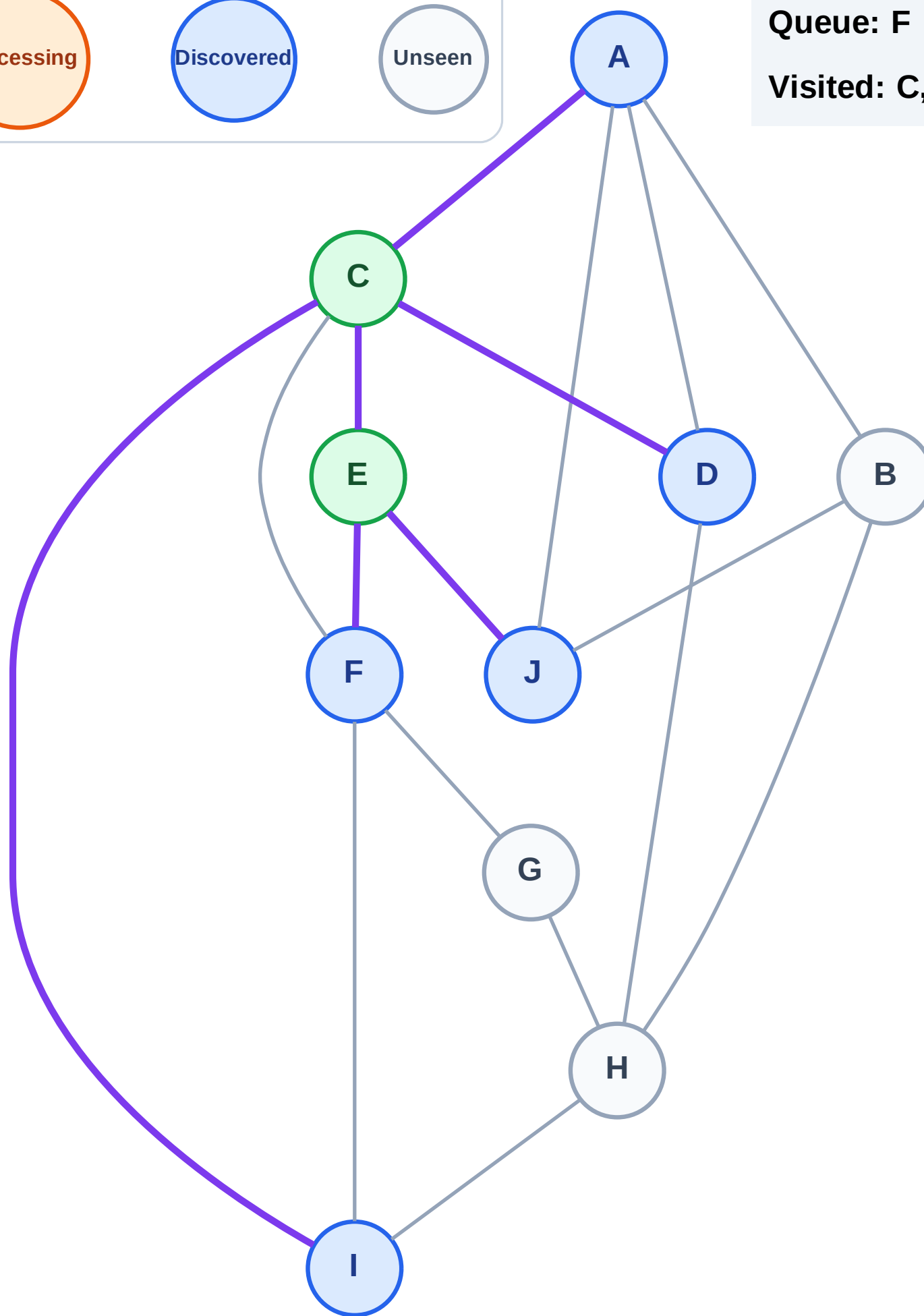
Discovered



Unseen

Queue: F → J → A → D → I

Visited: C, E



BFS Traversal

Step 6: Process node F

Legend

Complete

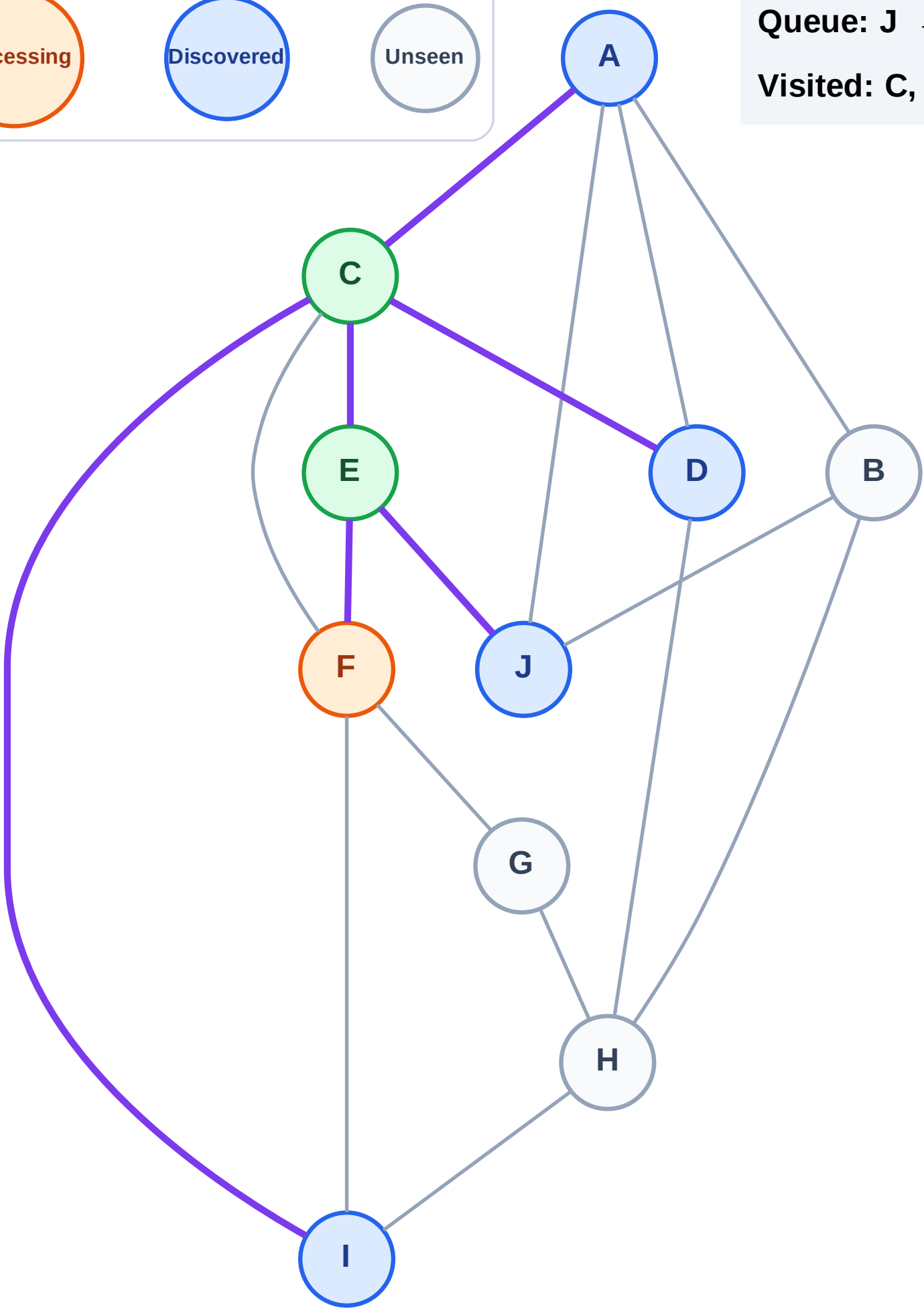
Processing

Discovered

Unseen

Queue: J → A → D → I

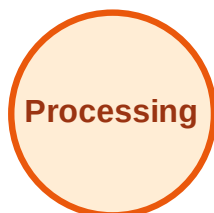
Visited: C, E



BFS Traversal

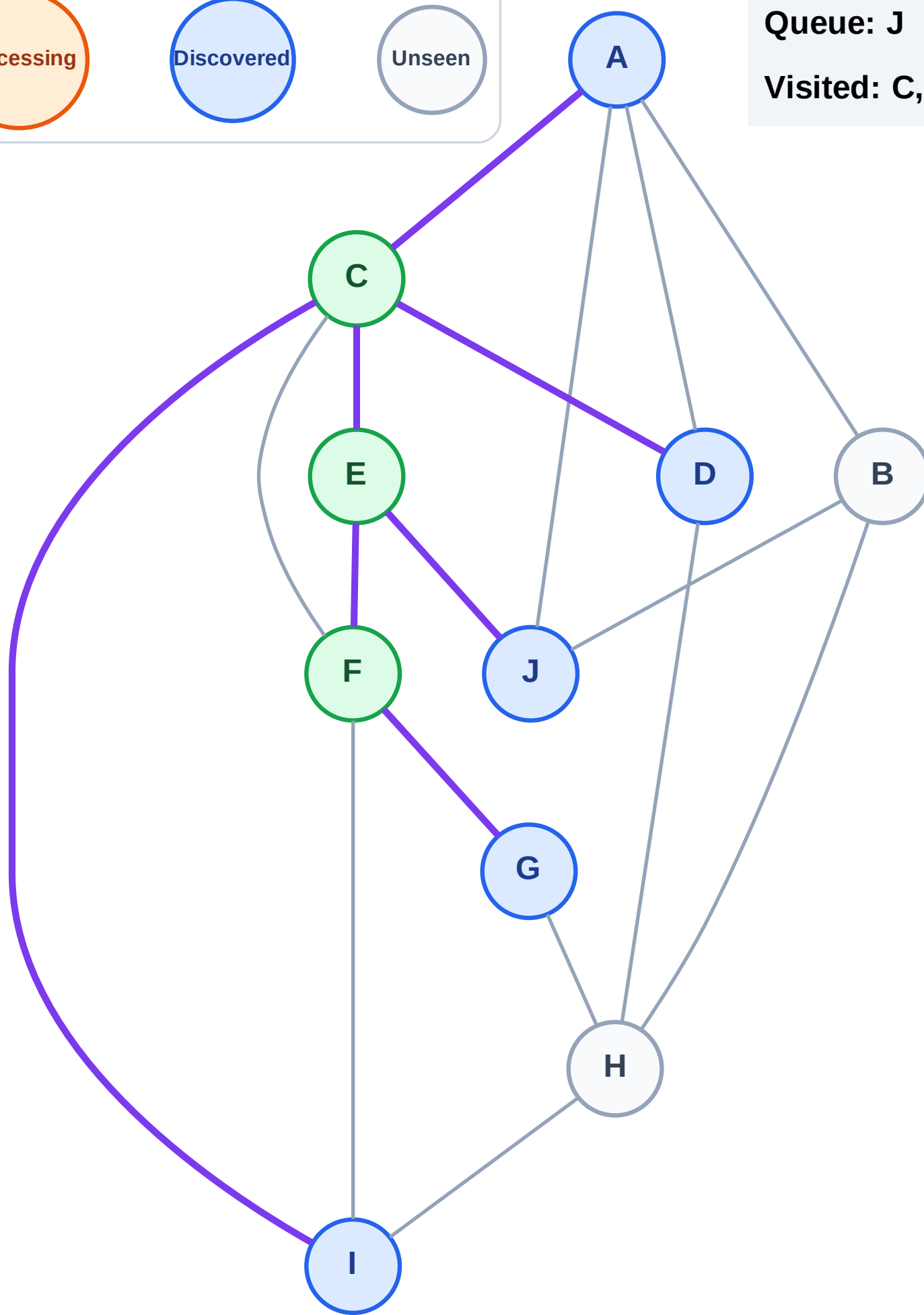
Step 7: Discover G from F

Legend



Queue: J → A → D → I → G

Visited: C, E, F



BFS Traversal

Step 8: Process node J

Legend

Complete

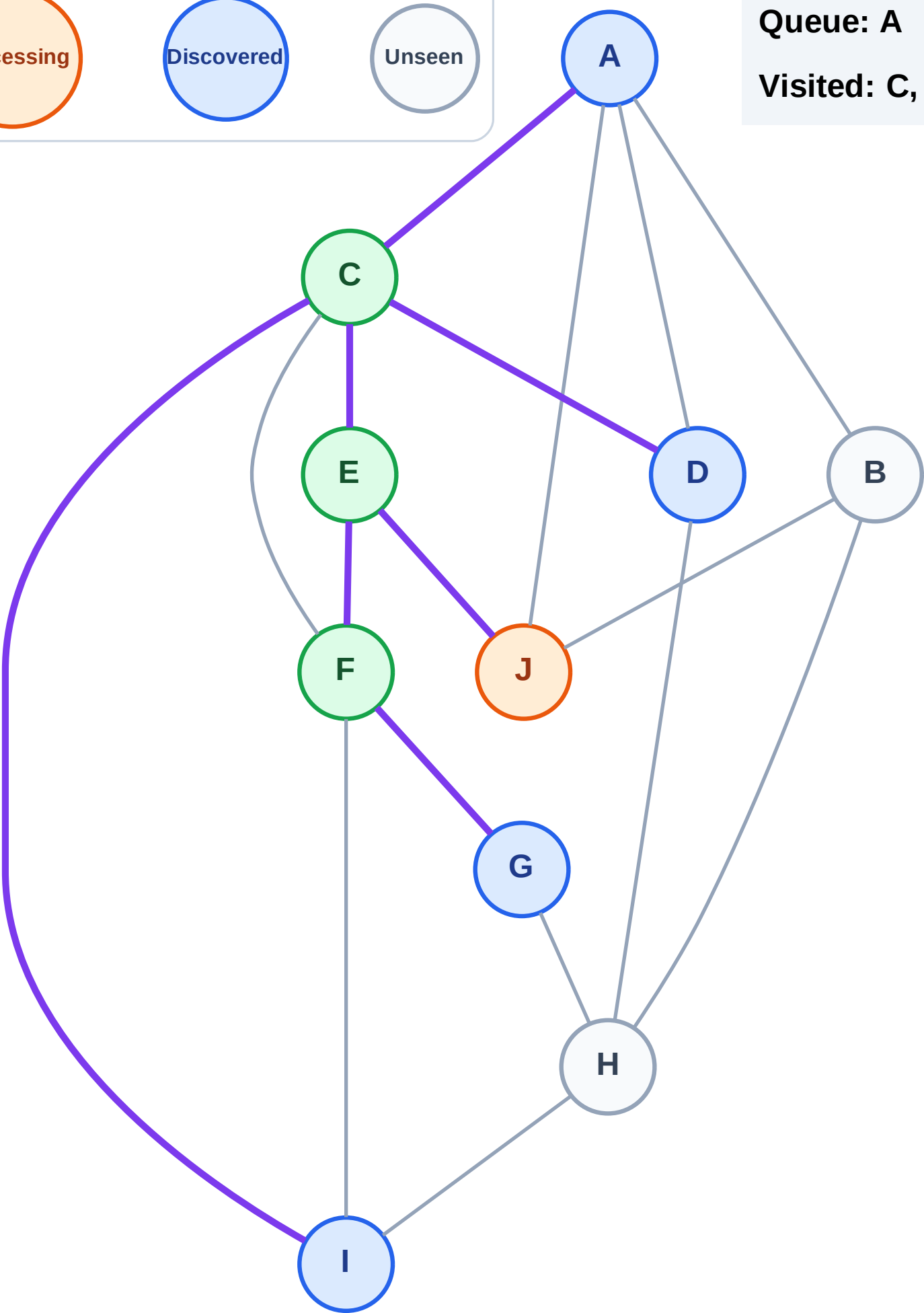
Processing

Discovered

Unseen

Queue: A → D → I → G

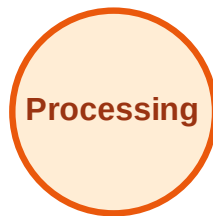
Visited: C, E, F



BFS Traversal

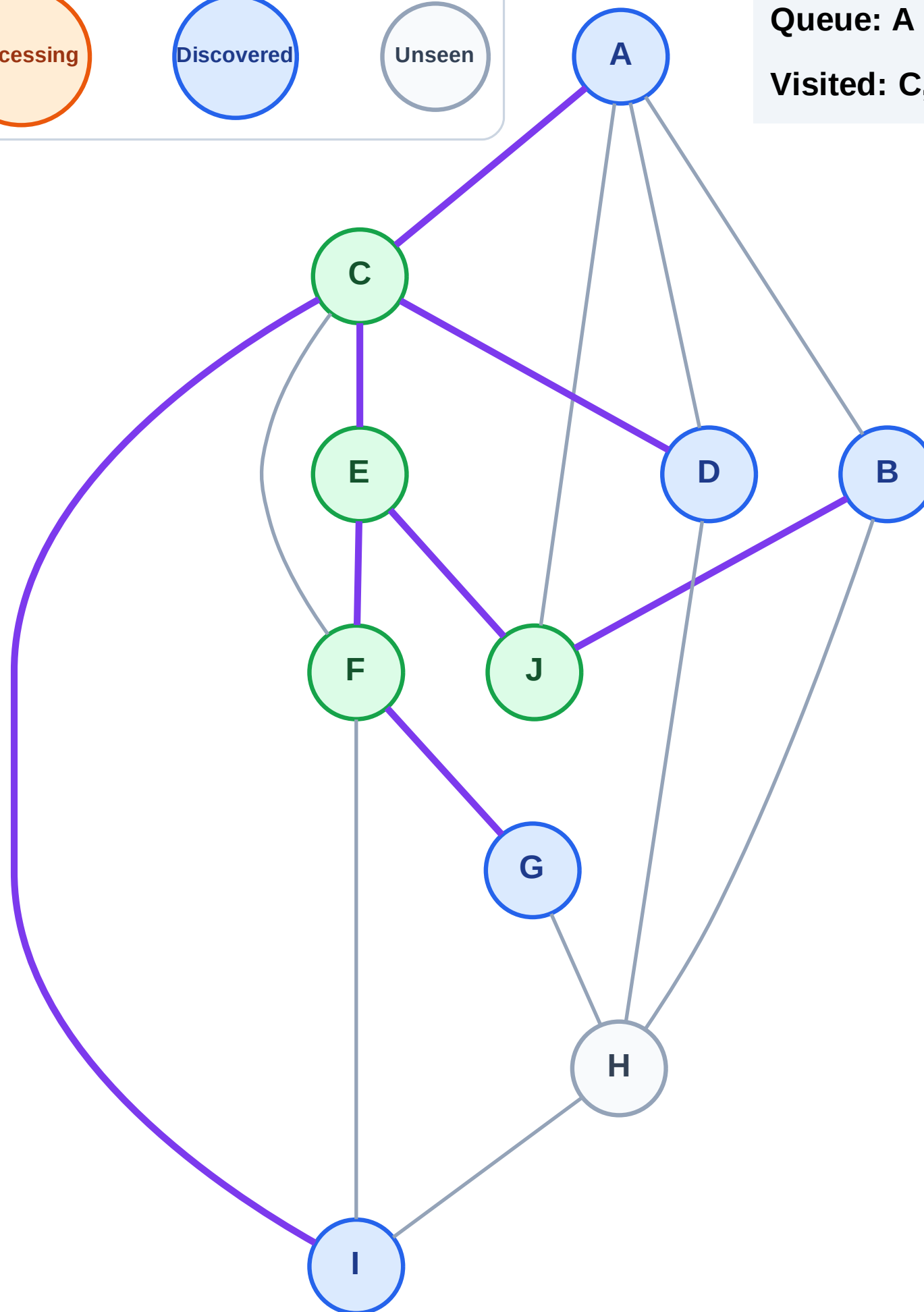
Step 9: Discover B from J

Legend



Queue: A → D → I → G → B

Visited: C, E, F, J



BFS Traversal

Step 10: Process node A

Legend

Complete

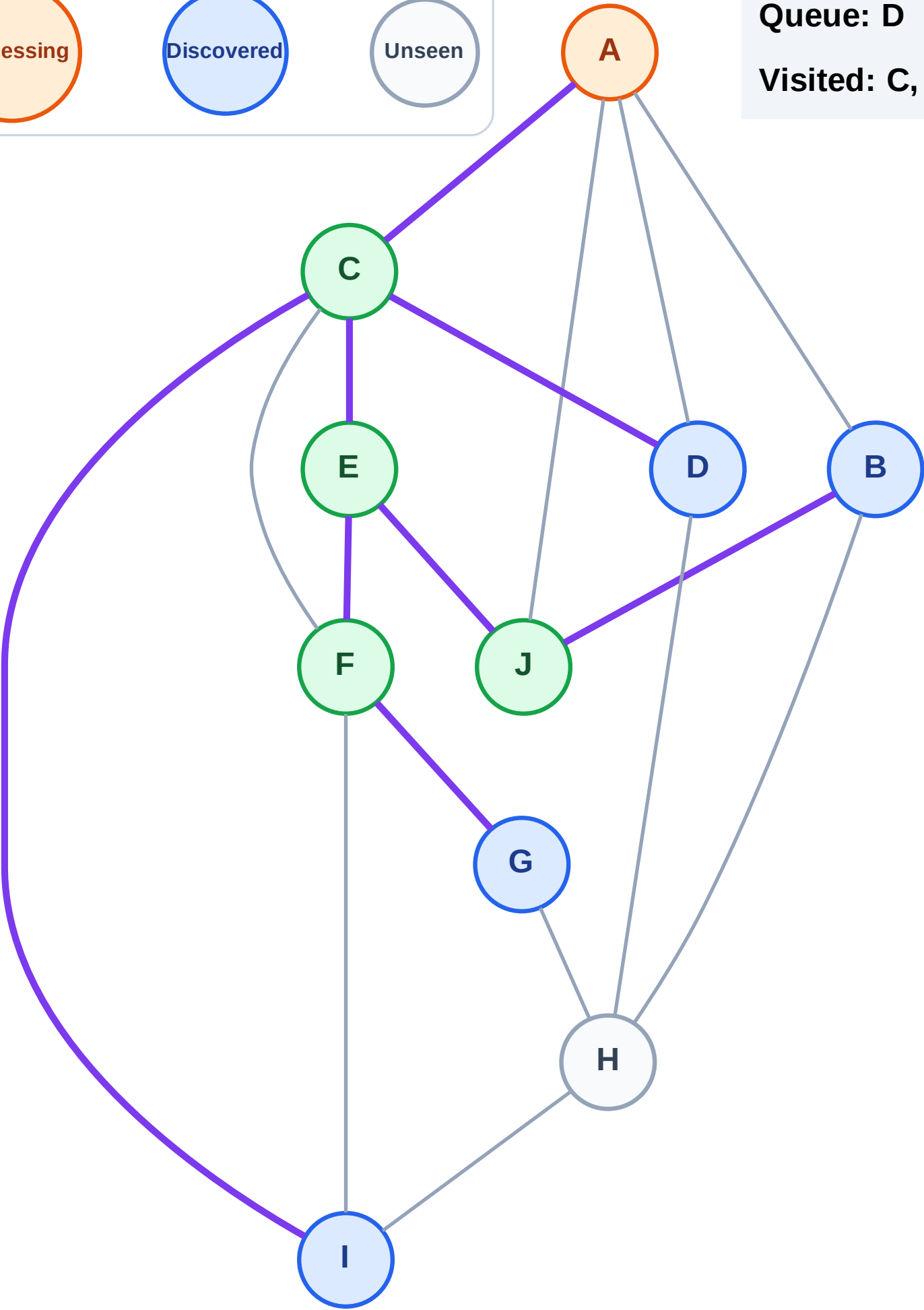
Processing

Discovered

Unseen

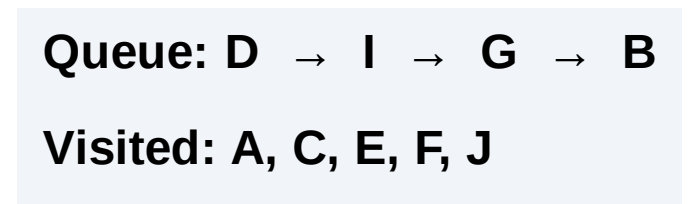
Queue: D → I → G → B

Visited: C, E, F, J



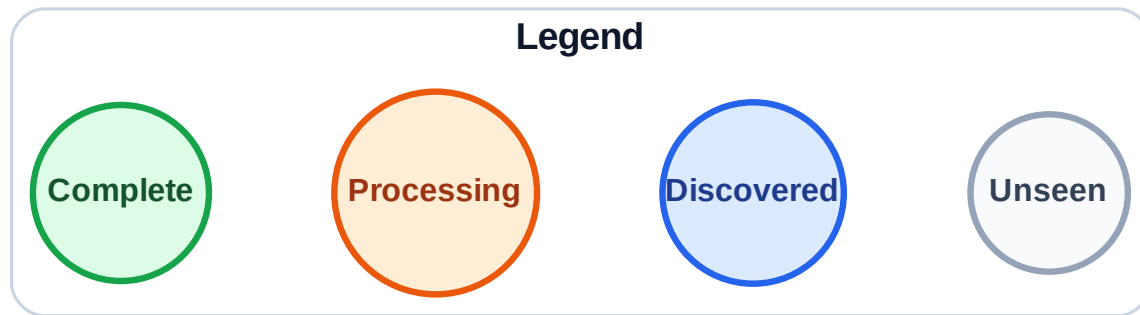
Step 11: No new neighbors from A

Step 11: No new neighbors from A

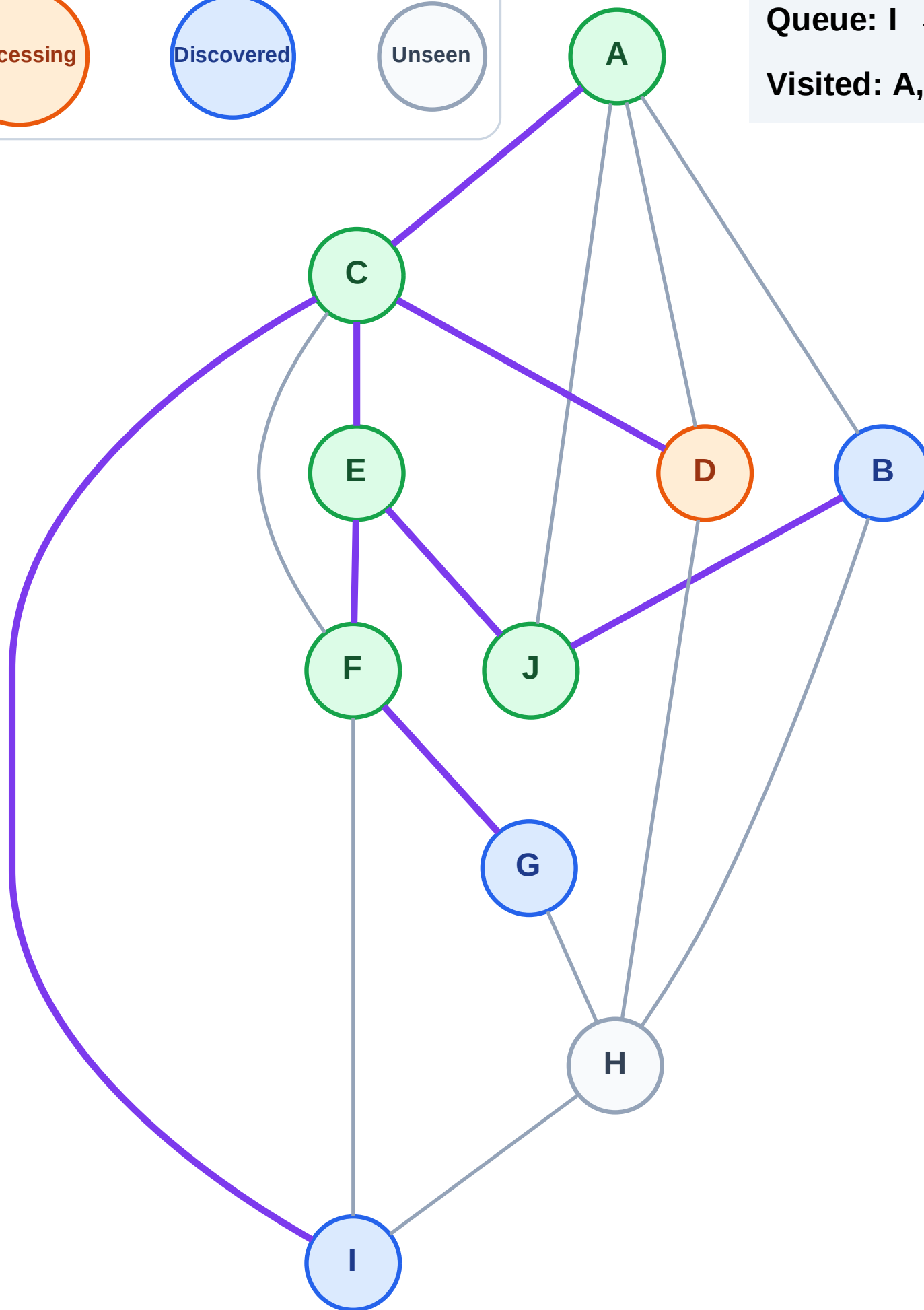


BFS Traversal

Step 12: Process node D

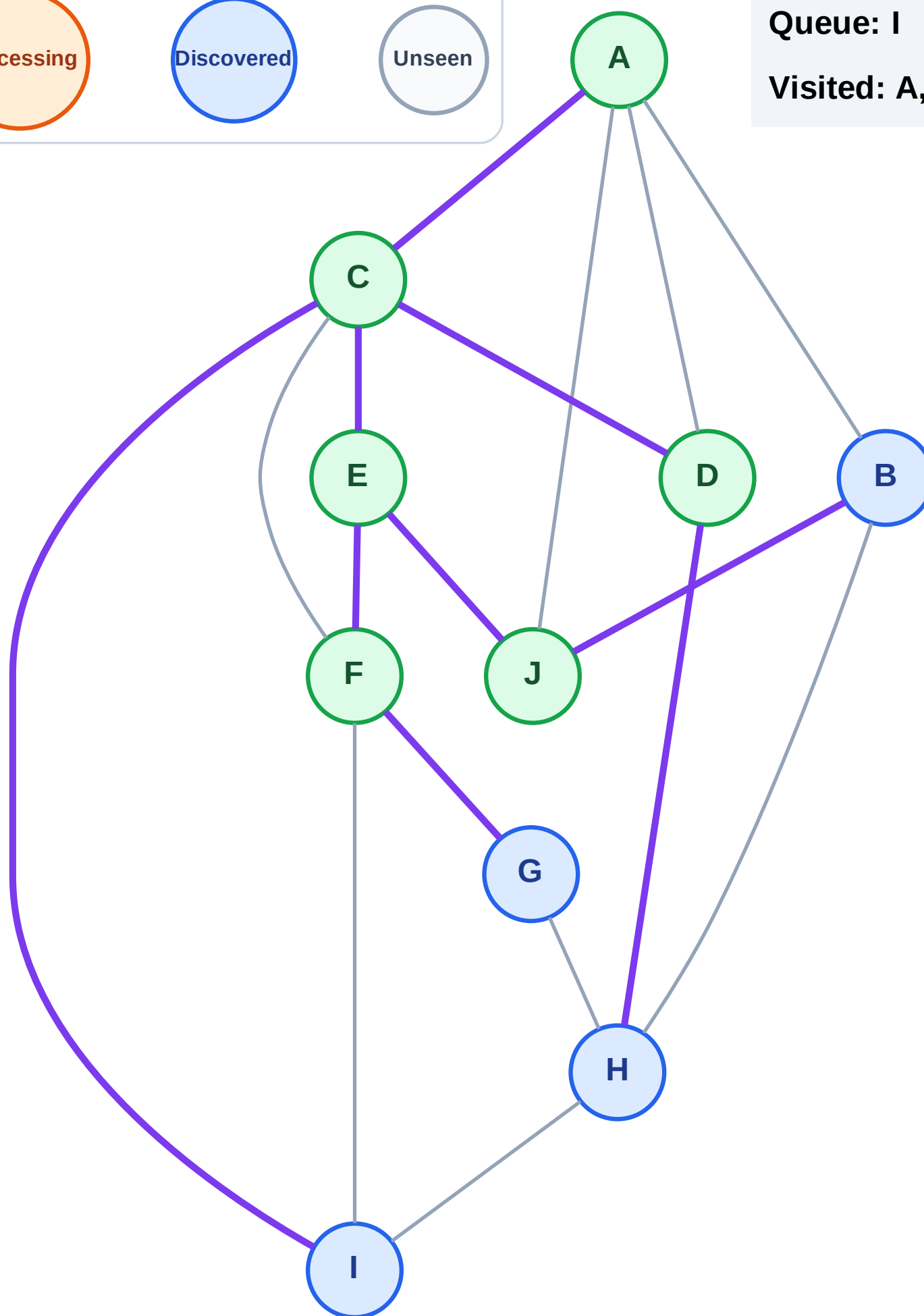
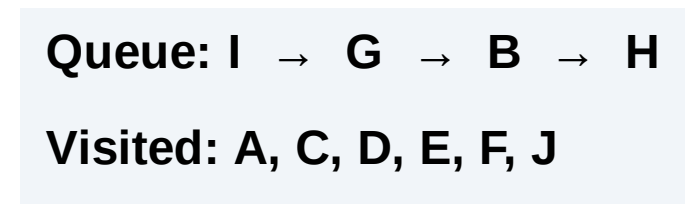


Queue: I → G → B
Visited: A, C, E, F, J

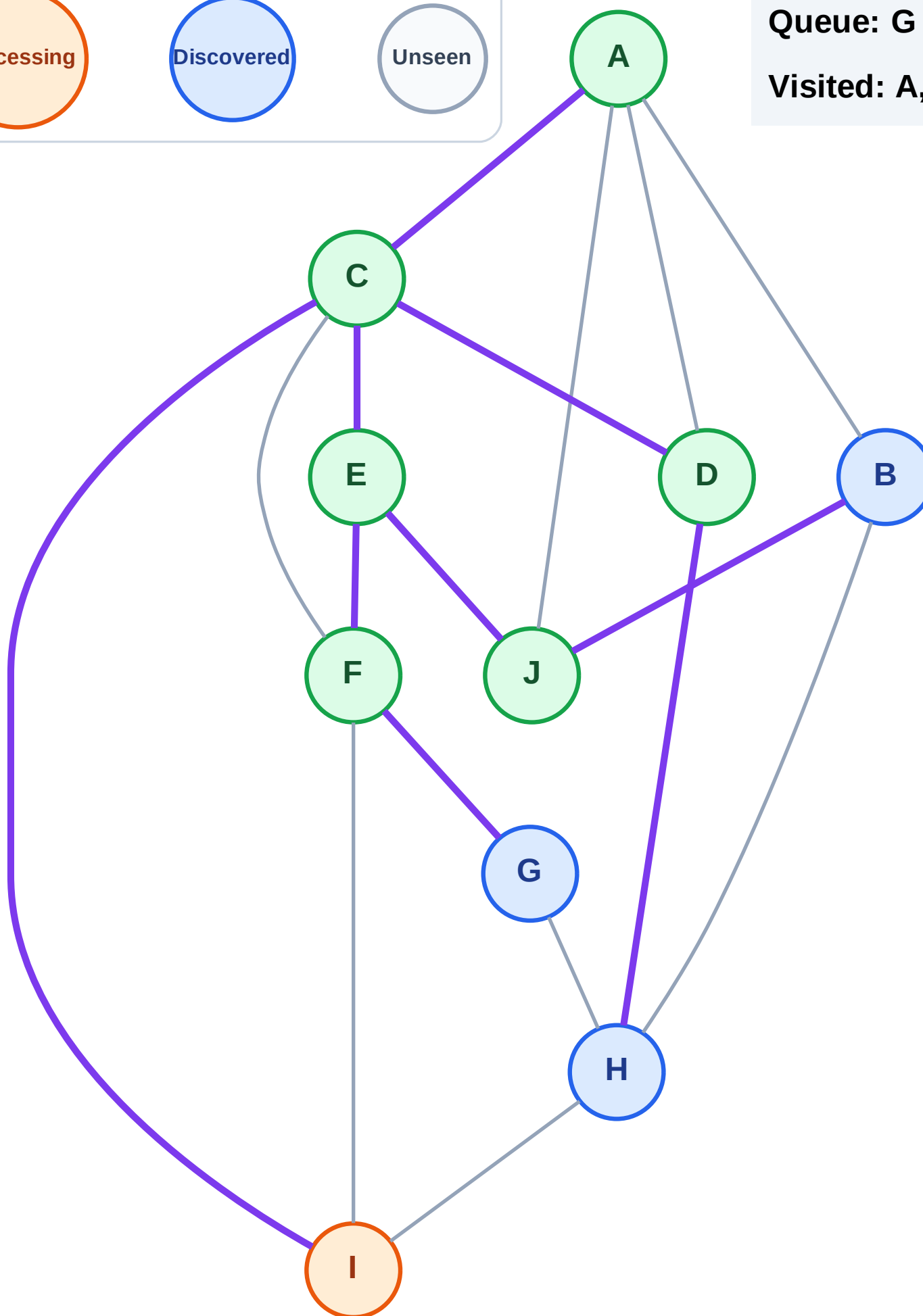
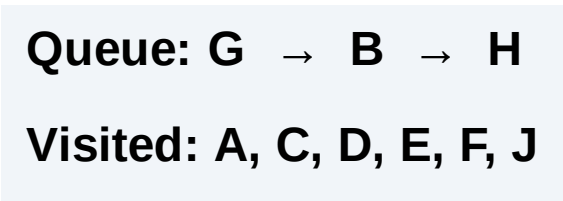


Step 13: Discover H from D

Step 13: Discover H from D



Step 14: Process node I

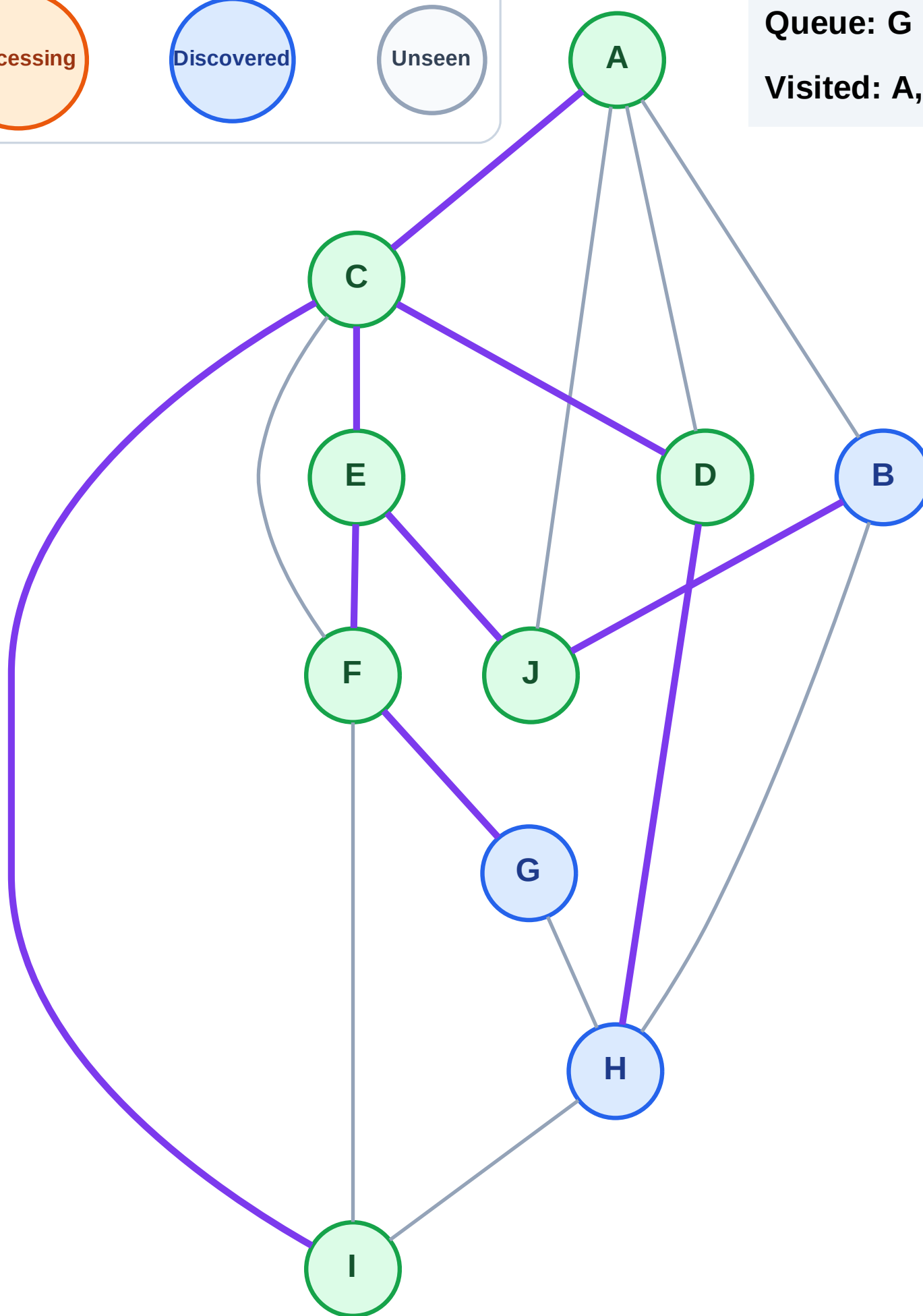


Step 15: No new neighbors from I

Legend

Unseen

Visited: A, C, D, E, F, I, J



BFS Traversal

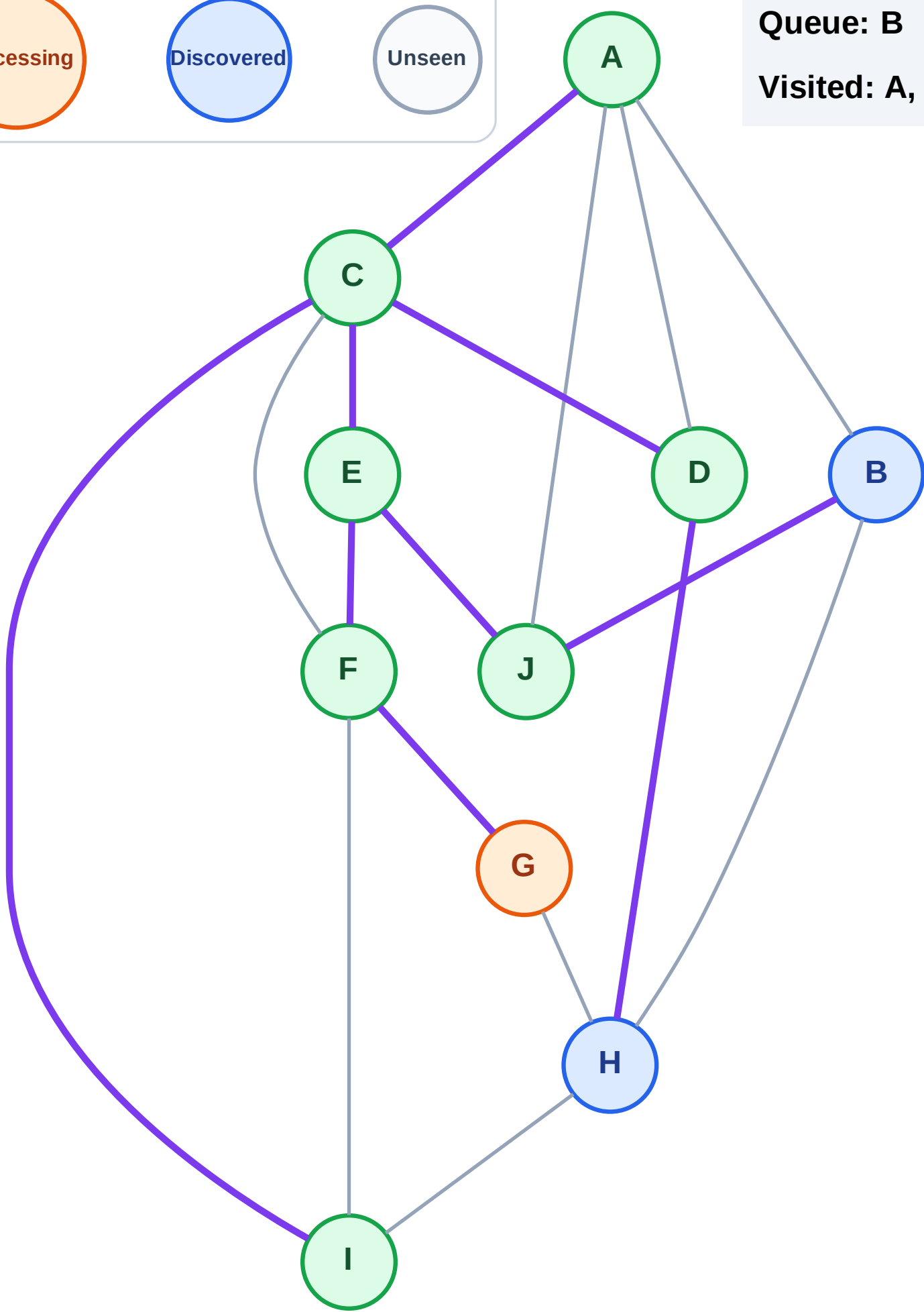
Step 16: Process node G

Legend



Queue: B → H

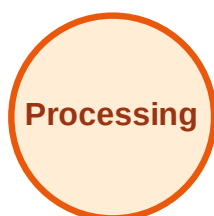
Visited: A, C, D, E, F, I, J



BFS Traversal

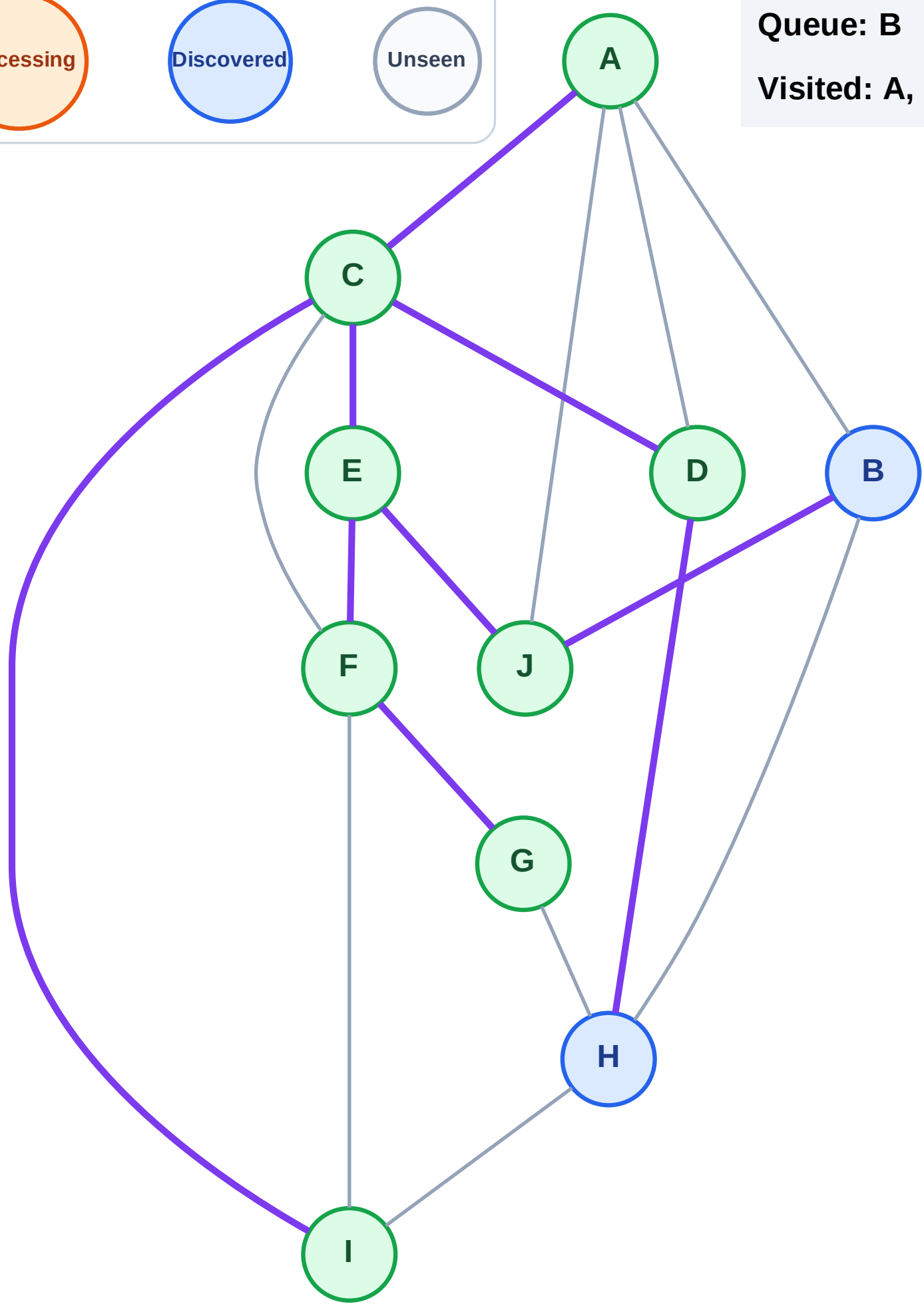
Step 17: No new neighbors from G

Legend



Queue: B → H

Visited: A, C, D, E, F, G, I, J



BFS Traversal

Step 18: Process node B

Legend

Complete

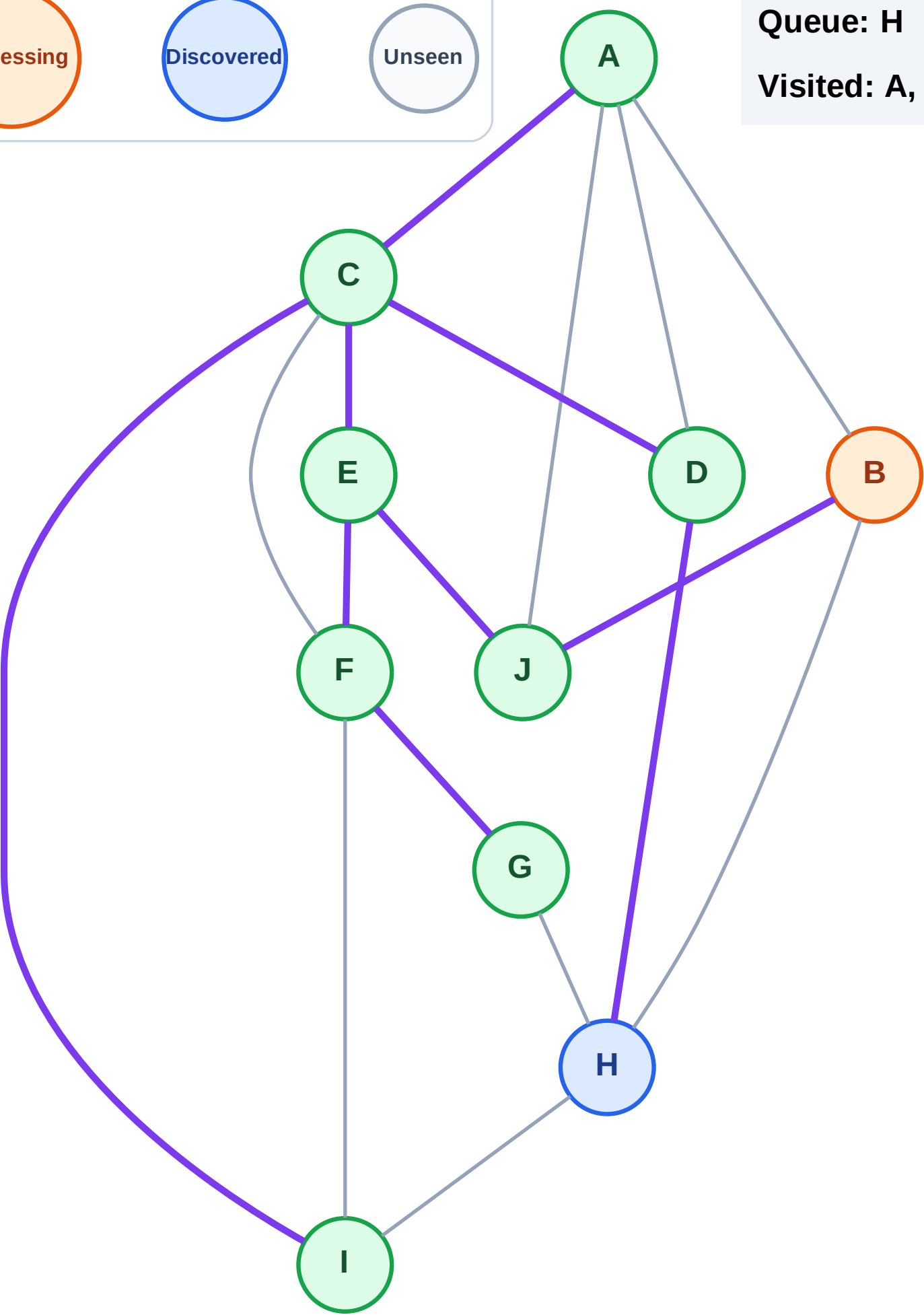
Processing

Discovered

Unseen

Queue: H

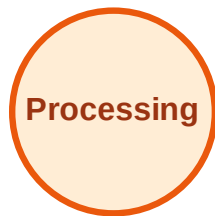
Visited: A, C, D, E, F, G, I, J



BFS Traversal

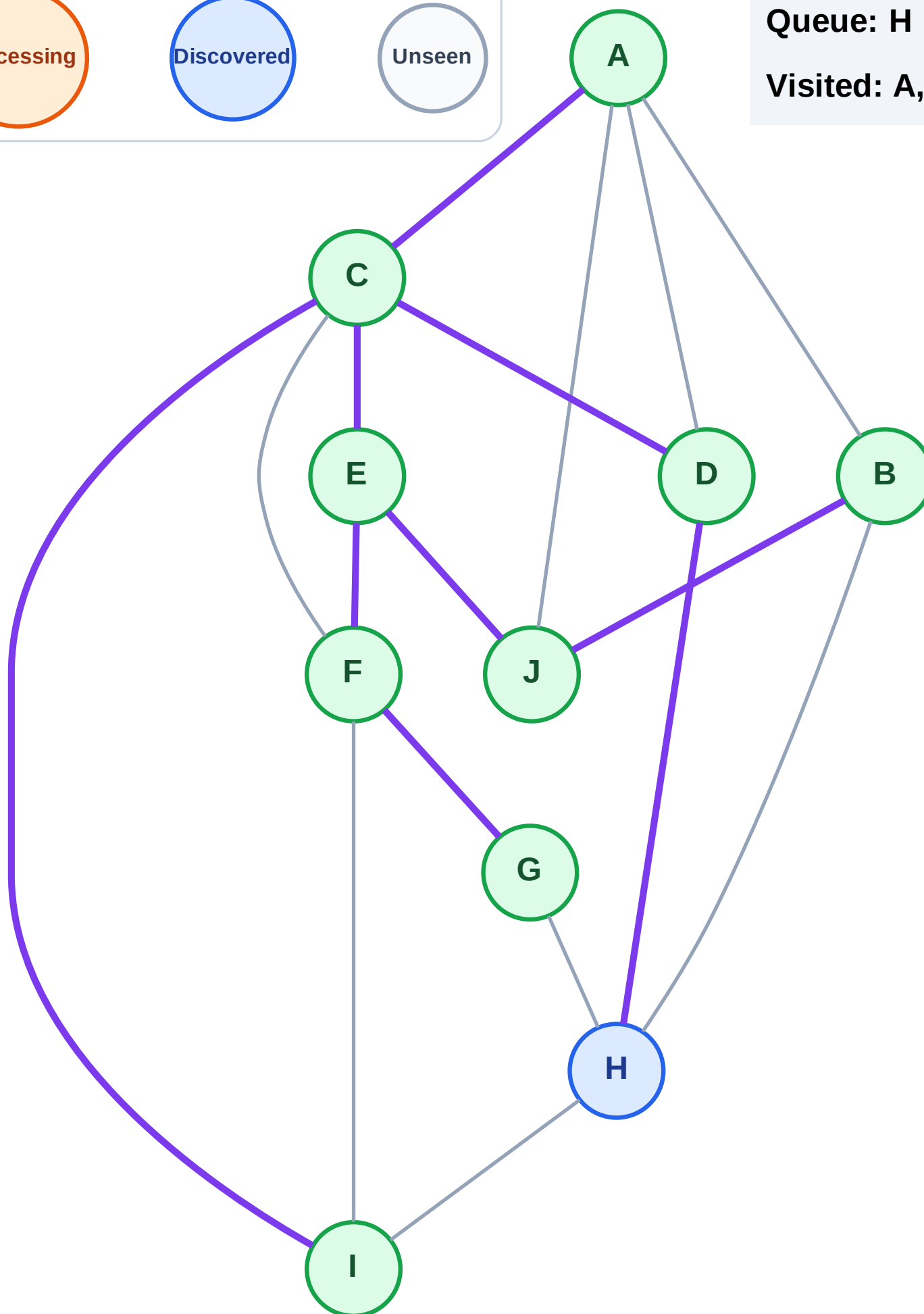
Step 19: No new neighbors from B

Legend



Queue: H

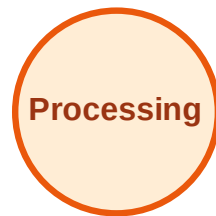
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

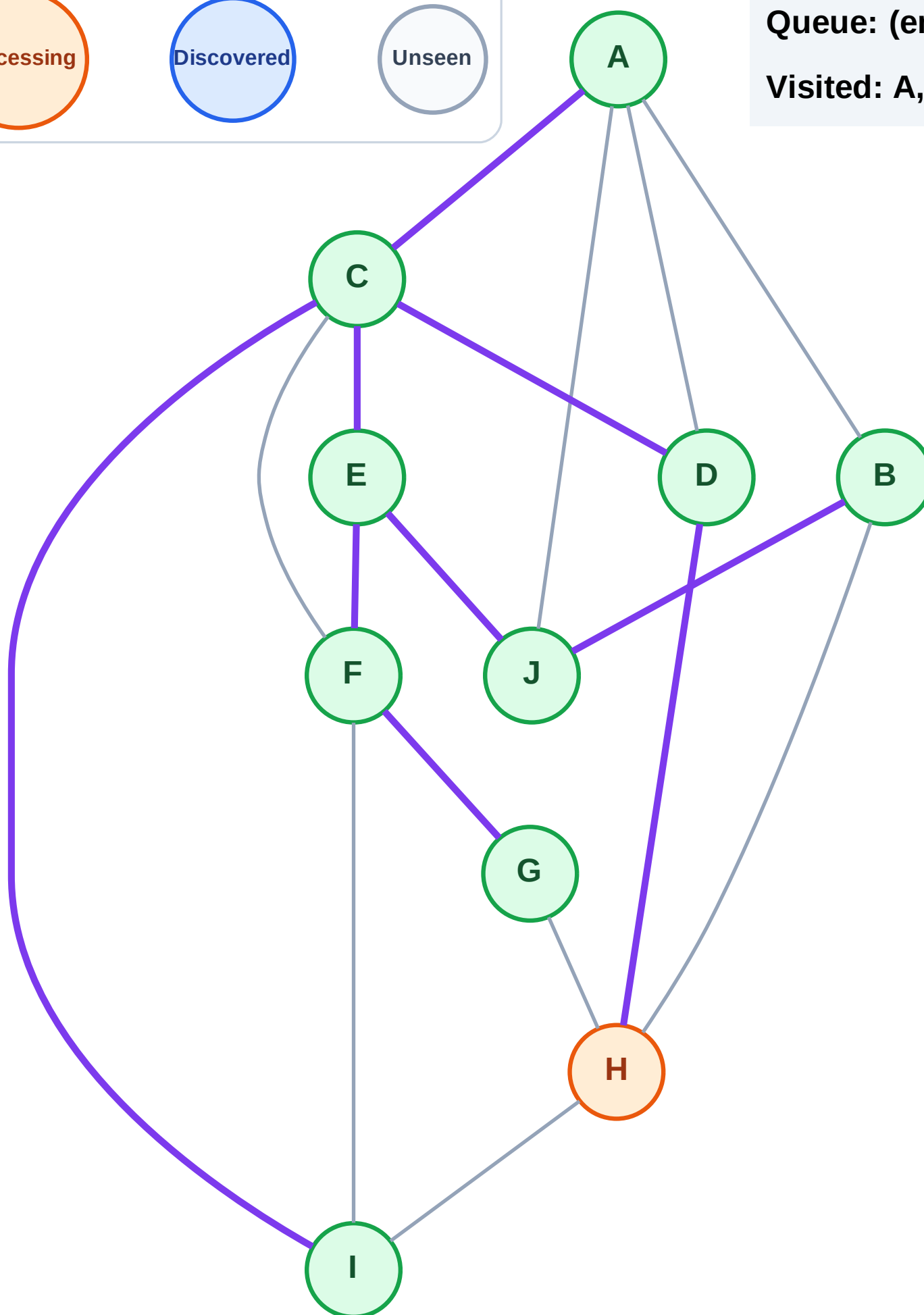
Step 20: Process node H

Legend



Queue: (empty)

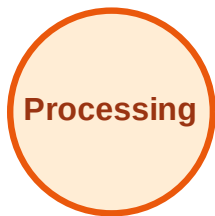
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

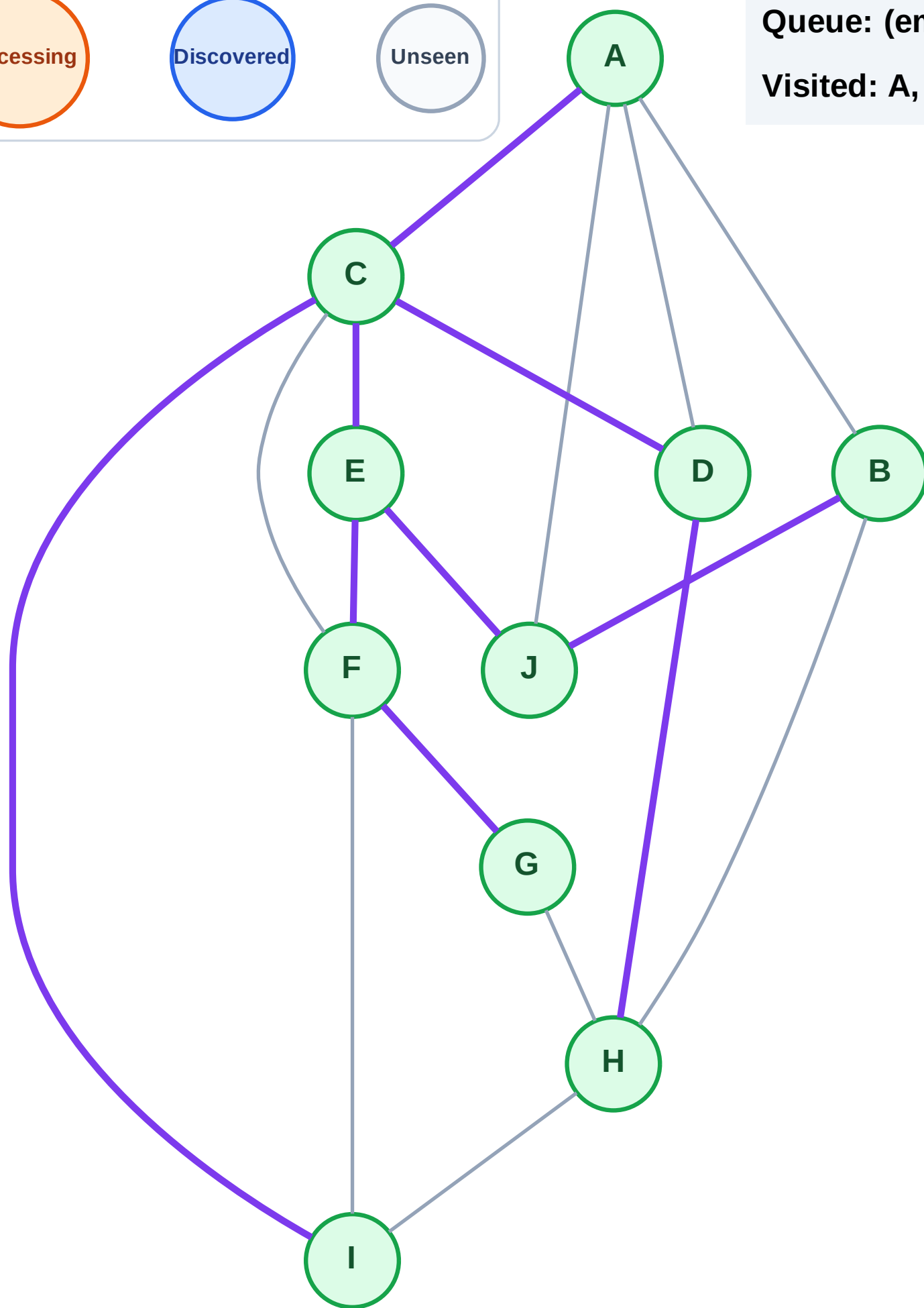
Step 21: No new neighbors from H

Legend



Queue: (empty)

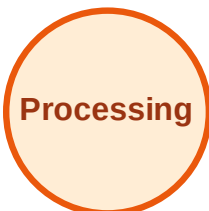
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

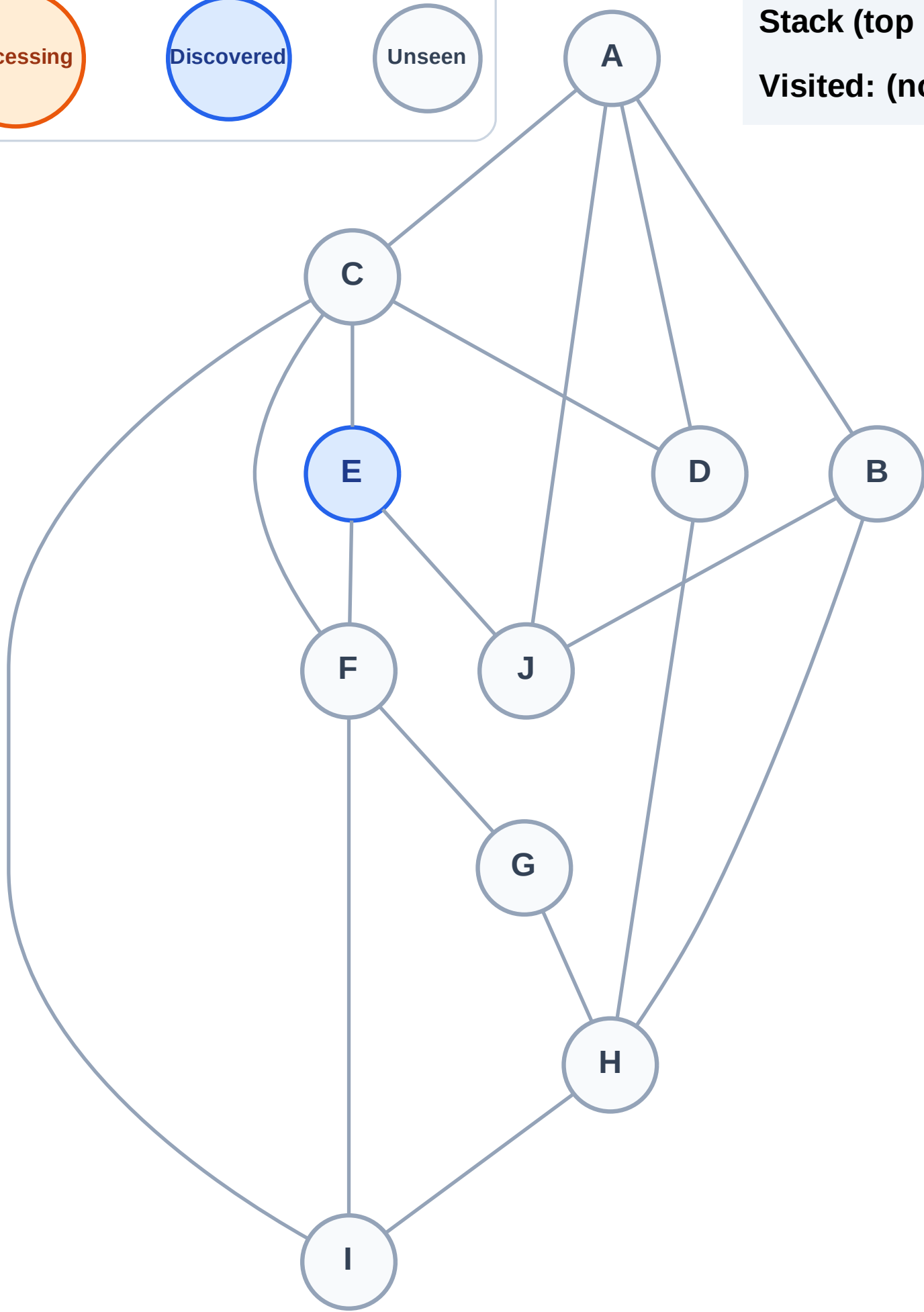
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

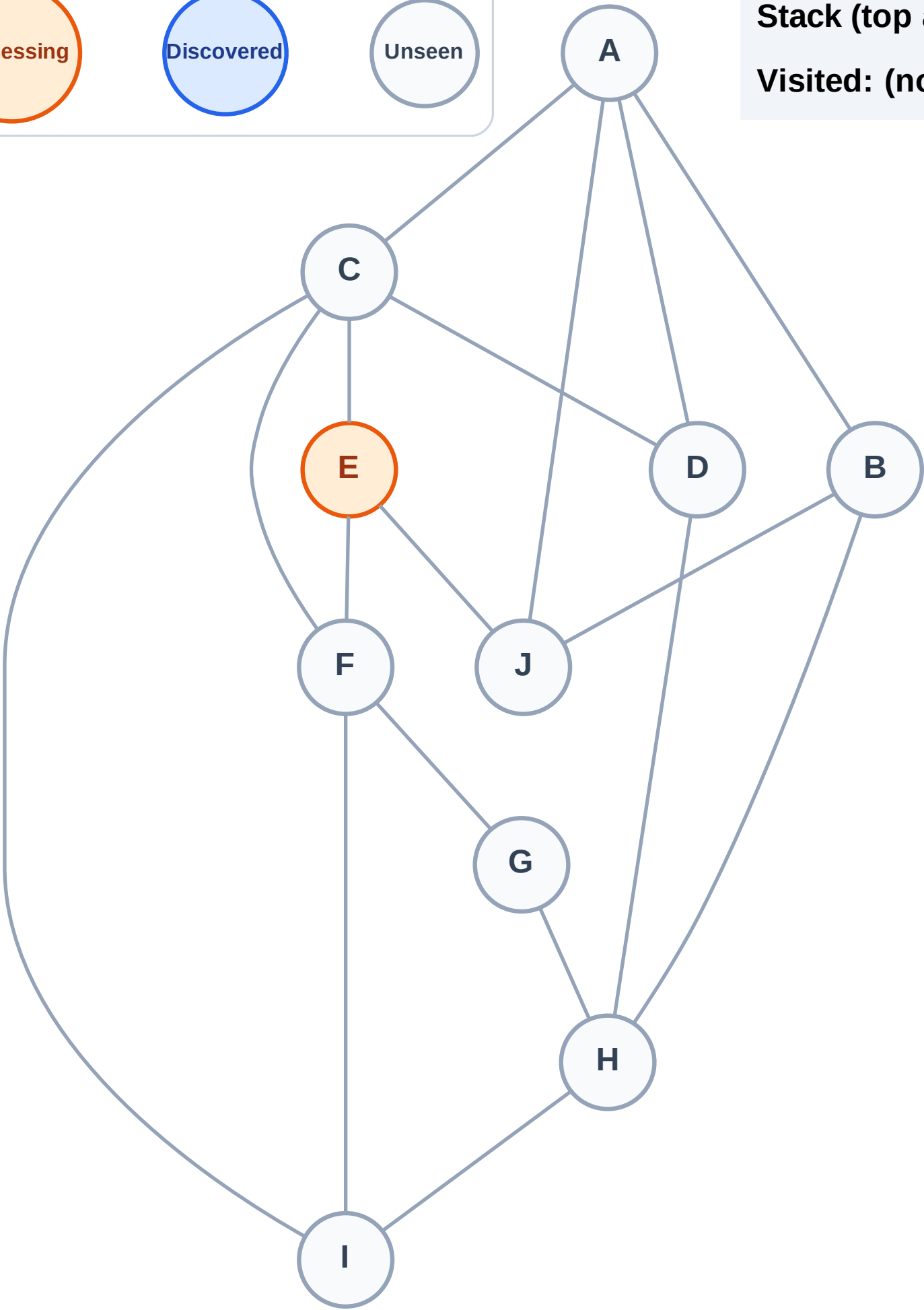
Complete

Processing

Discovered

Unseen

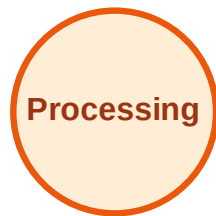
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

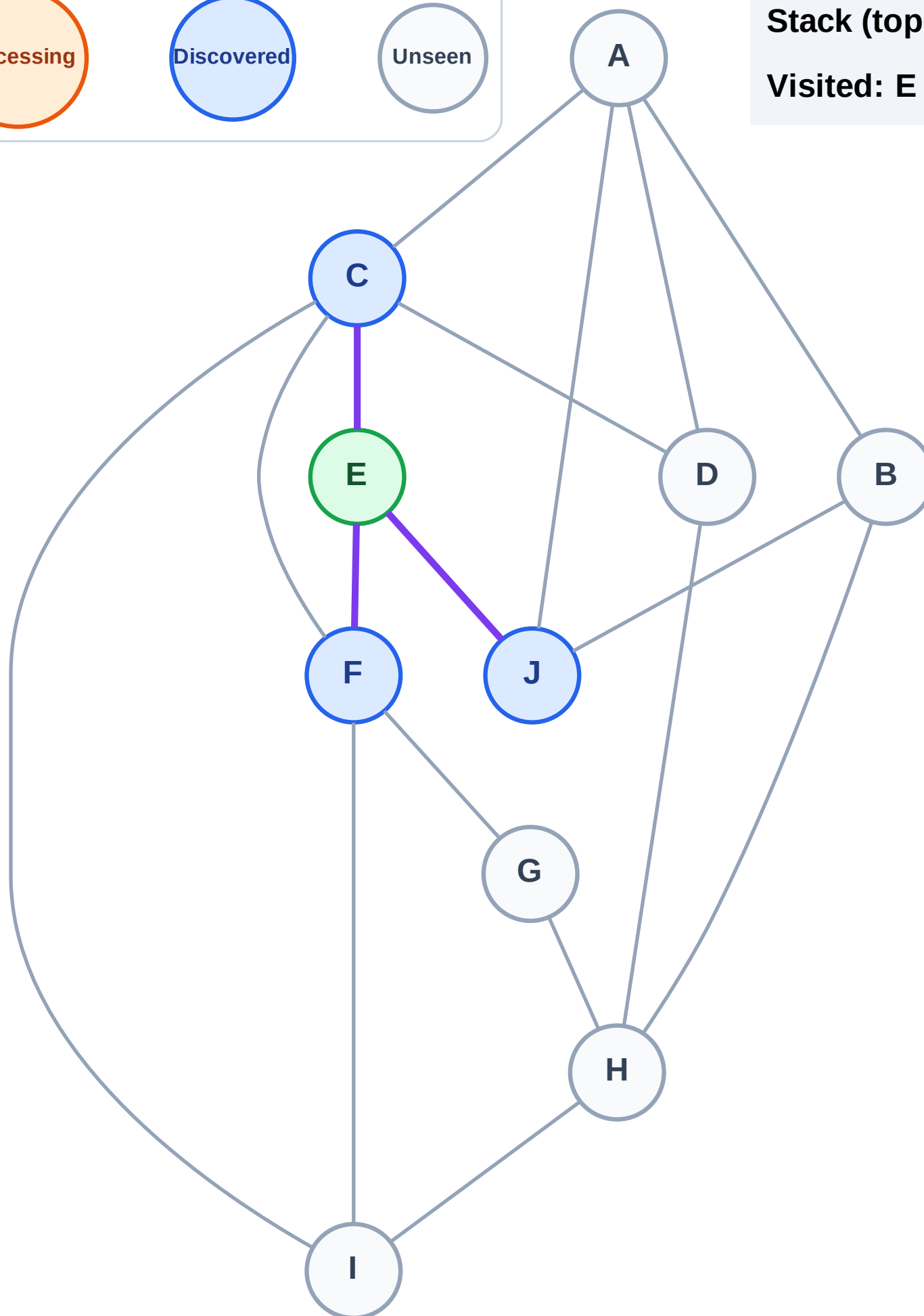
Step 3: Discover J, F, C from E

Legend



Stack (top at right): J | F | C

Visited: E



DFS Traversal

Step 4: Process node C

Legend

Complete

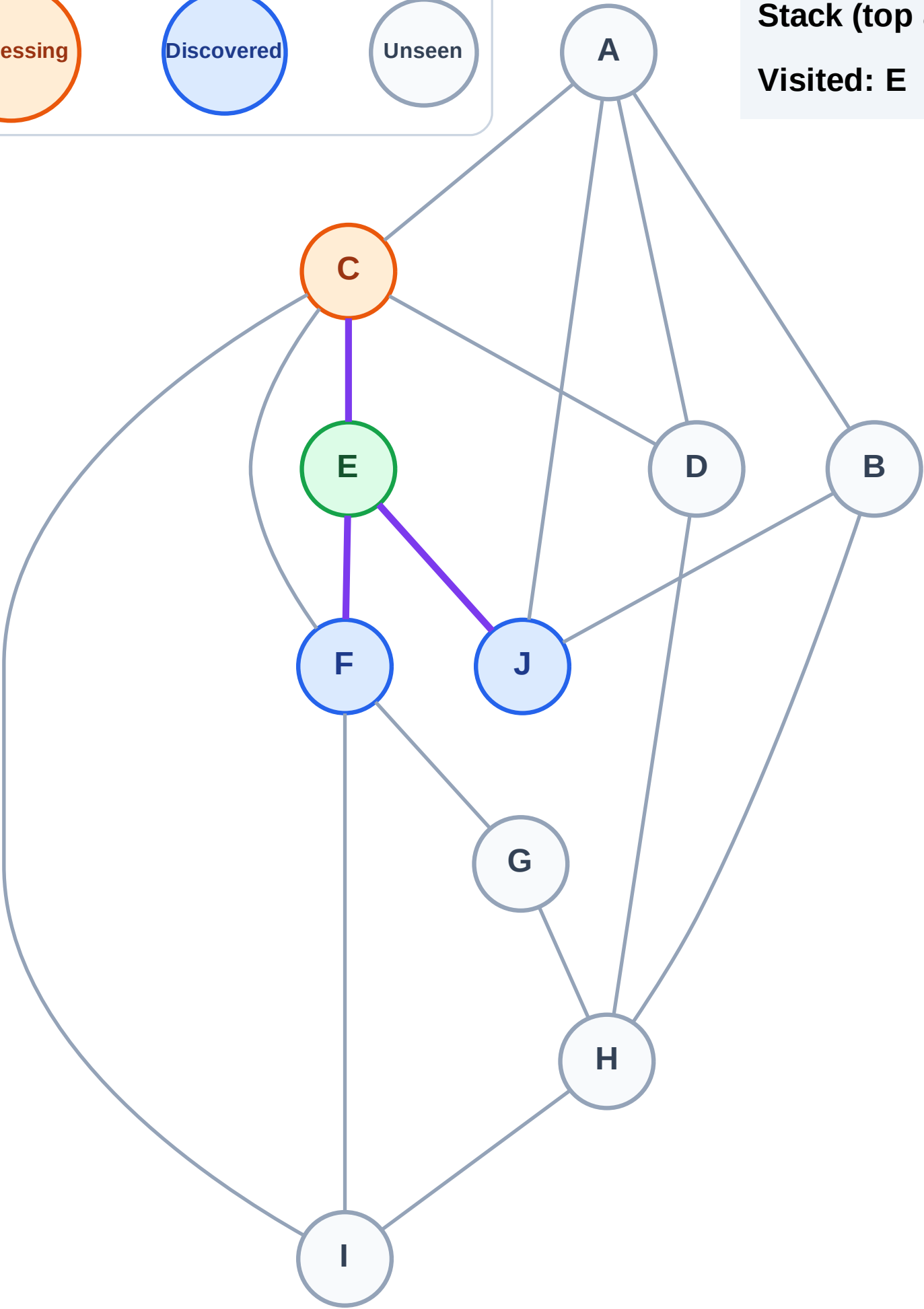
Processing

Discovered

Unseen

Stack (top at right): J | F

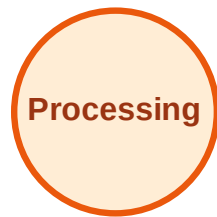
Visited: E



DFS Traversal

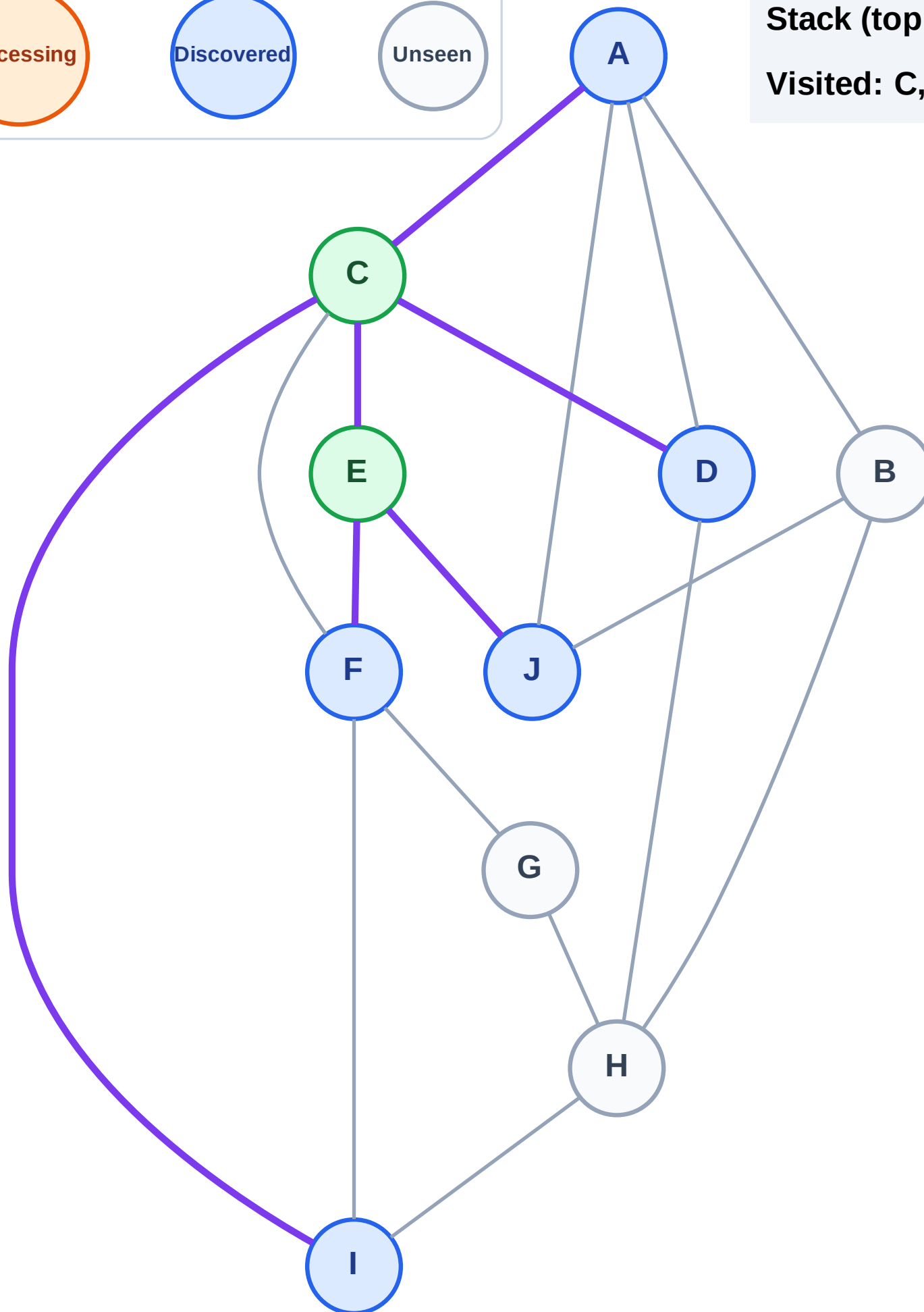
Step 5: Discover I, D, A from C

Legend



Stack (top at right): J | F | I | D | A

Visited: C, E



DFS Traversal

Step 6: Process node A

Legend



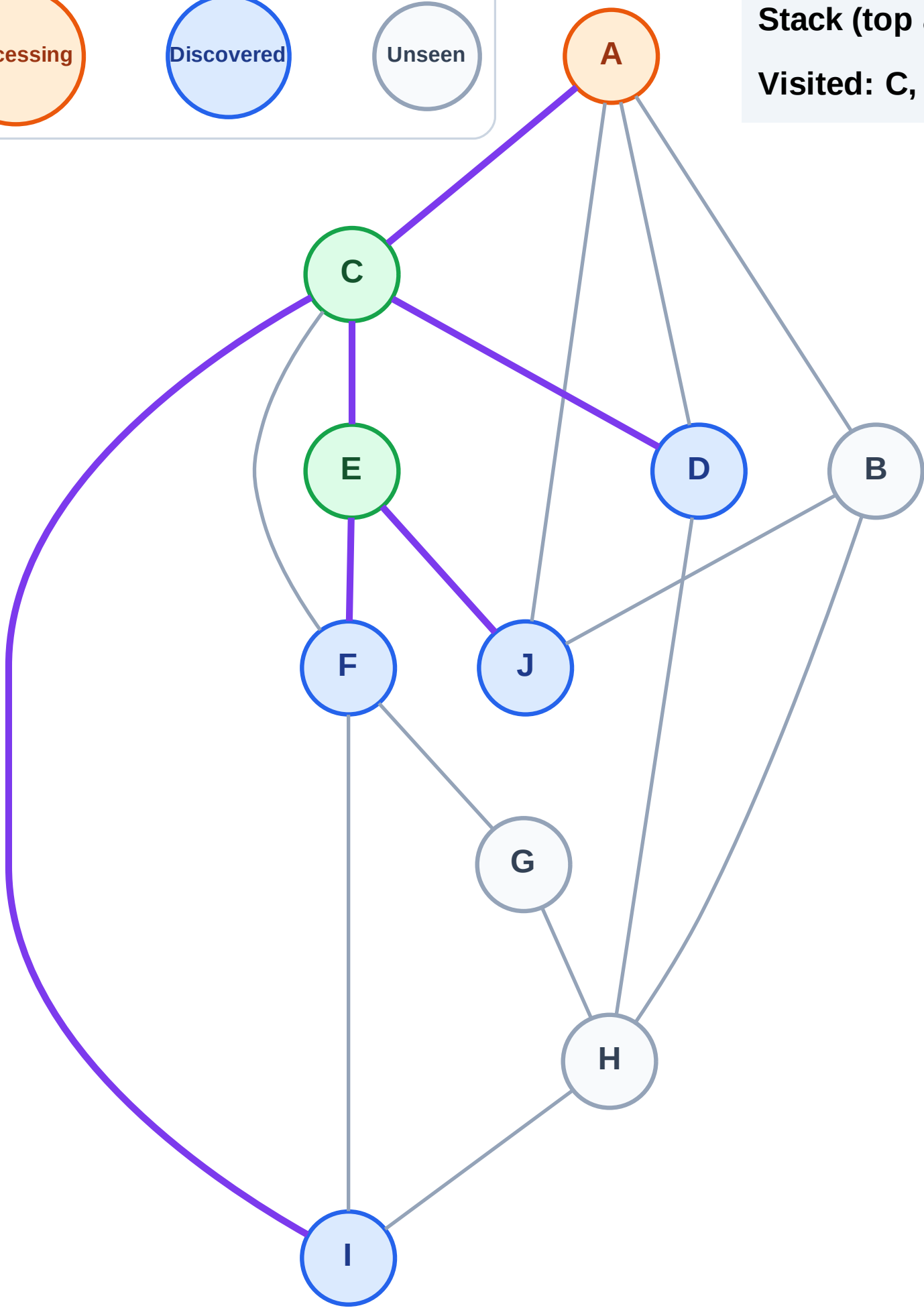
Processing



Unseen

Stack (top at right): J | F | I | D

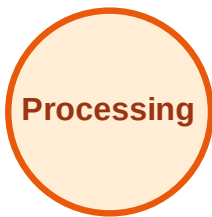
Visited: C, E



DFS Traversal

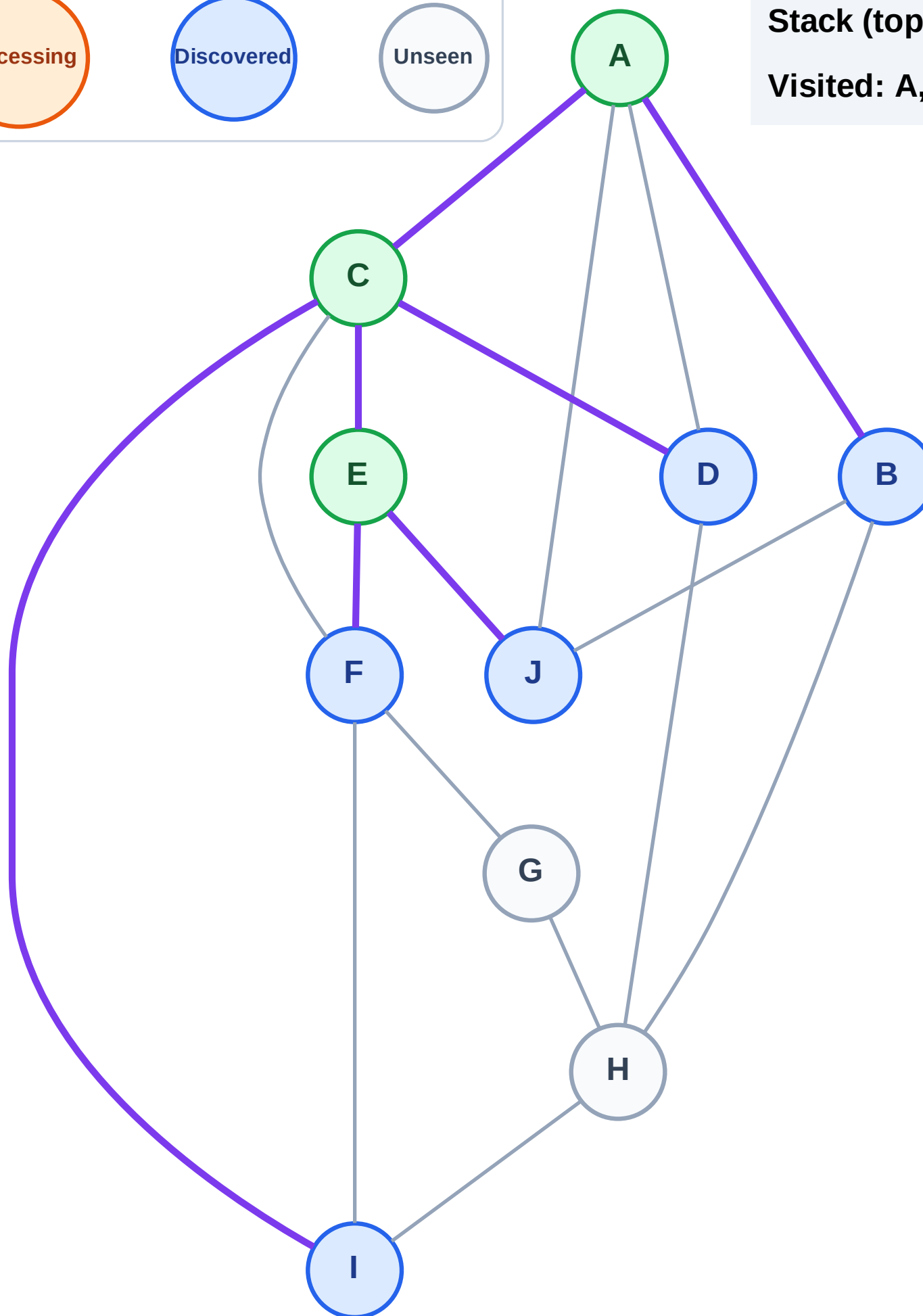
Step 7: Discover B from A

Legend



Stack (top at right): J | F | I | D | B

Visited: A, C, E



DFS Traversal

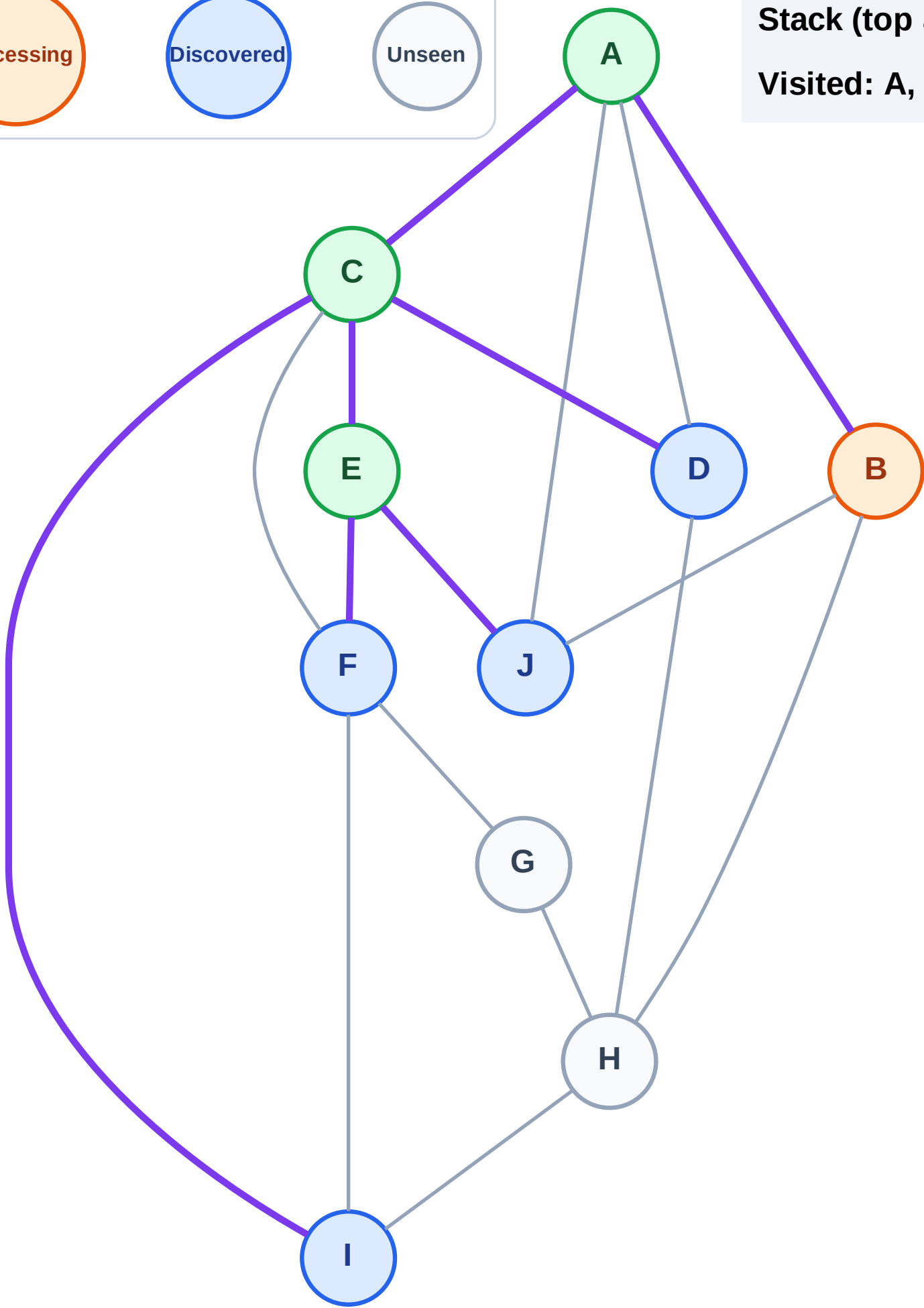
Step 8: Process node B

Legend



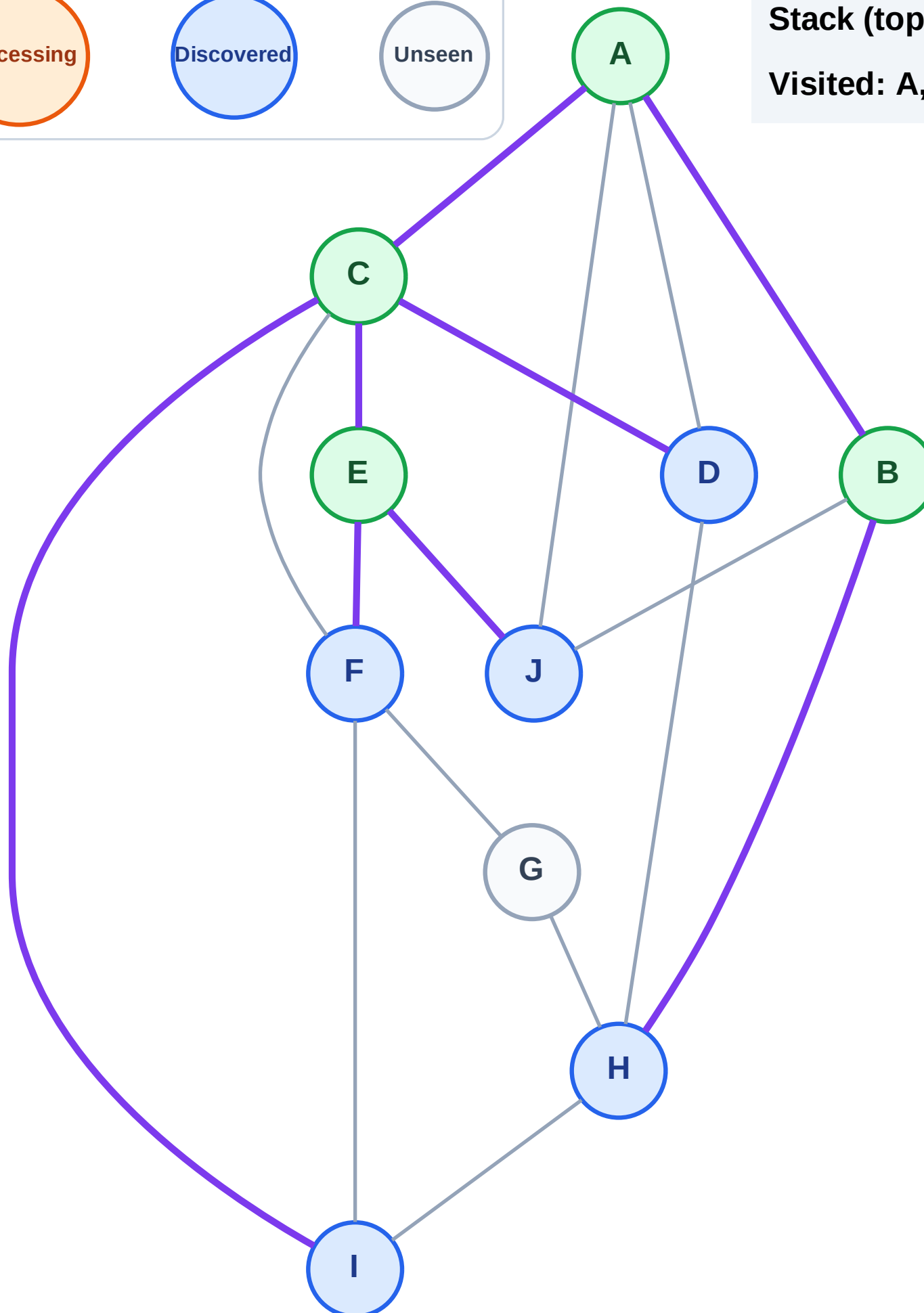
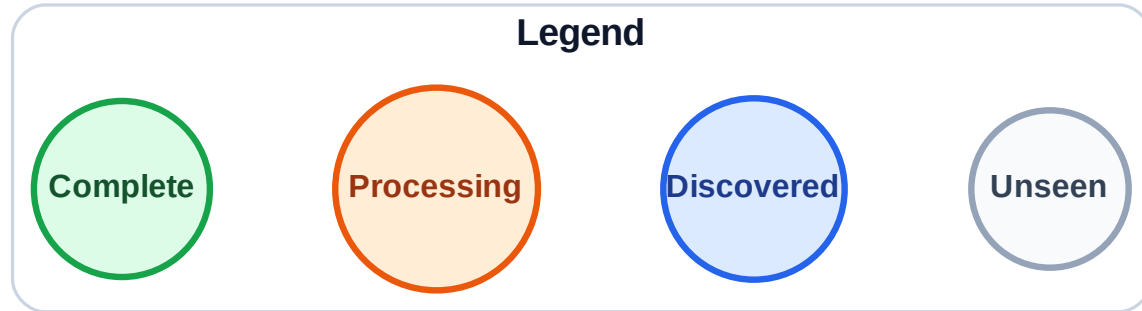
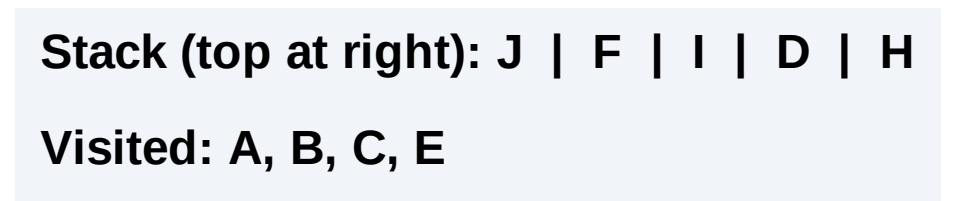
Stack (top at right): J | F | I | D

Visited: A, C, E



Step 9: Discover H from B

Step 9: Discover H from B



DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



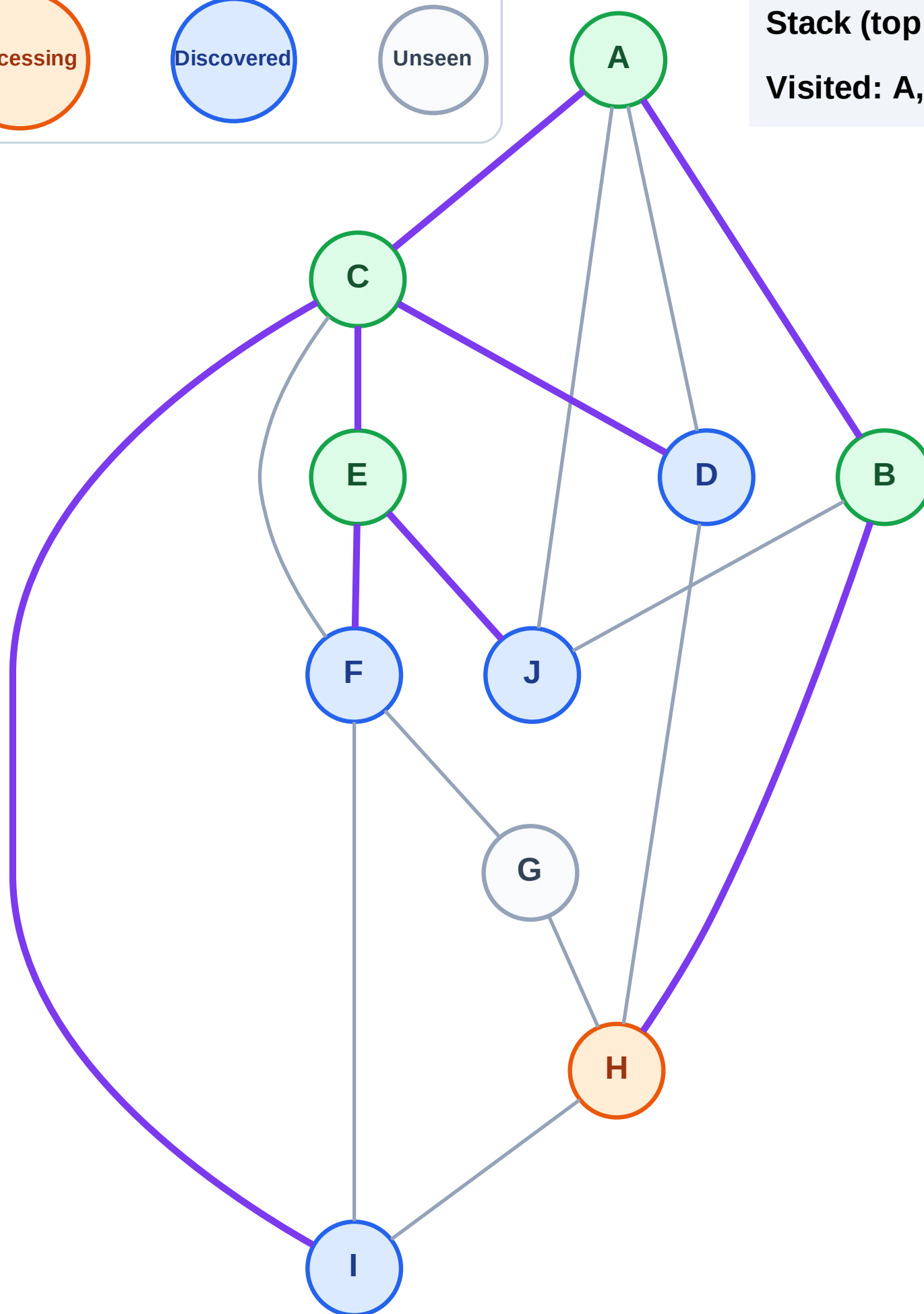
Discovered



Unseen

Stack (top at right): J | F | I | D

Visited: A, B, C, E



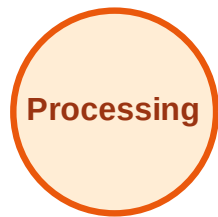
DFS Traversal

Step 11: Discover G from H

Legend



Complete



Processing



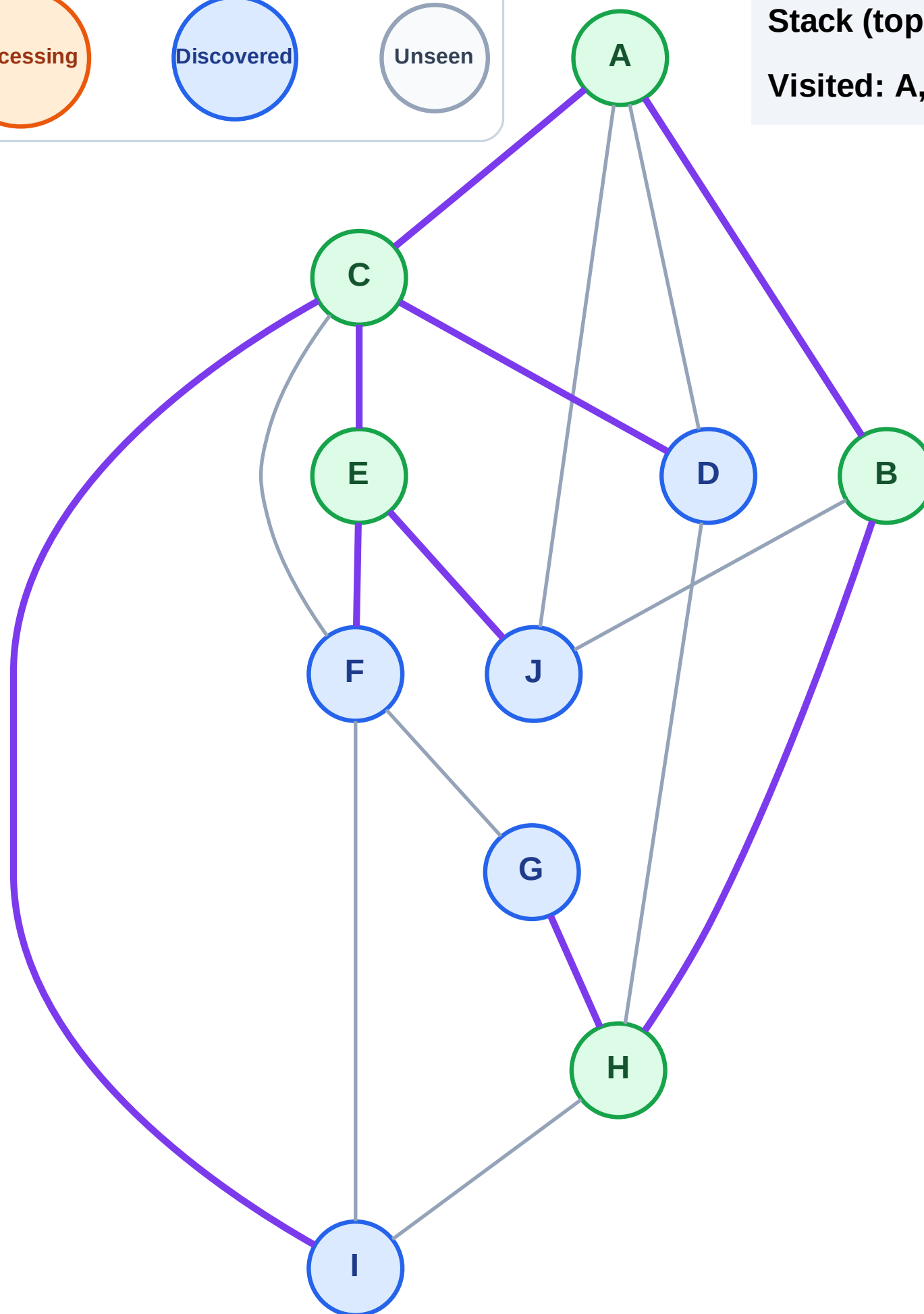
Discovered



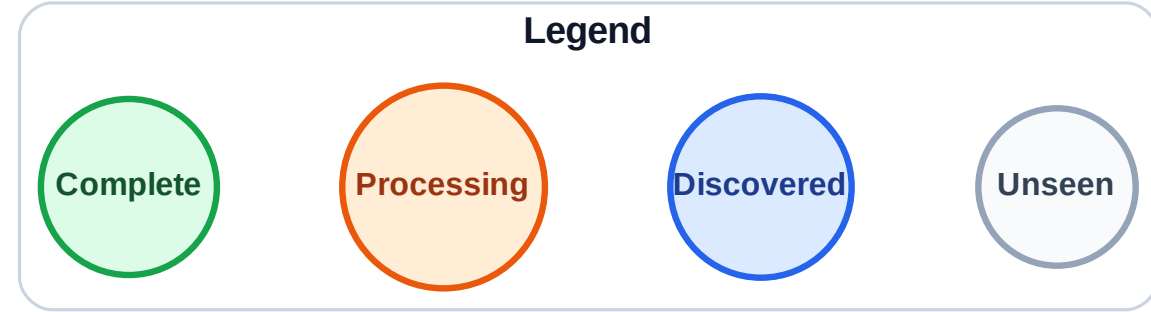
Unseen

Stack (top at right): J | F | I | D | G

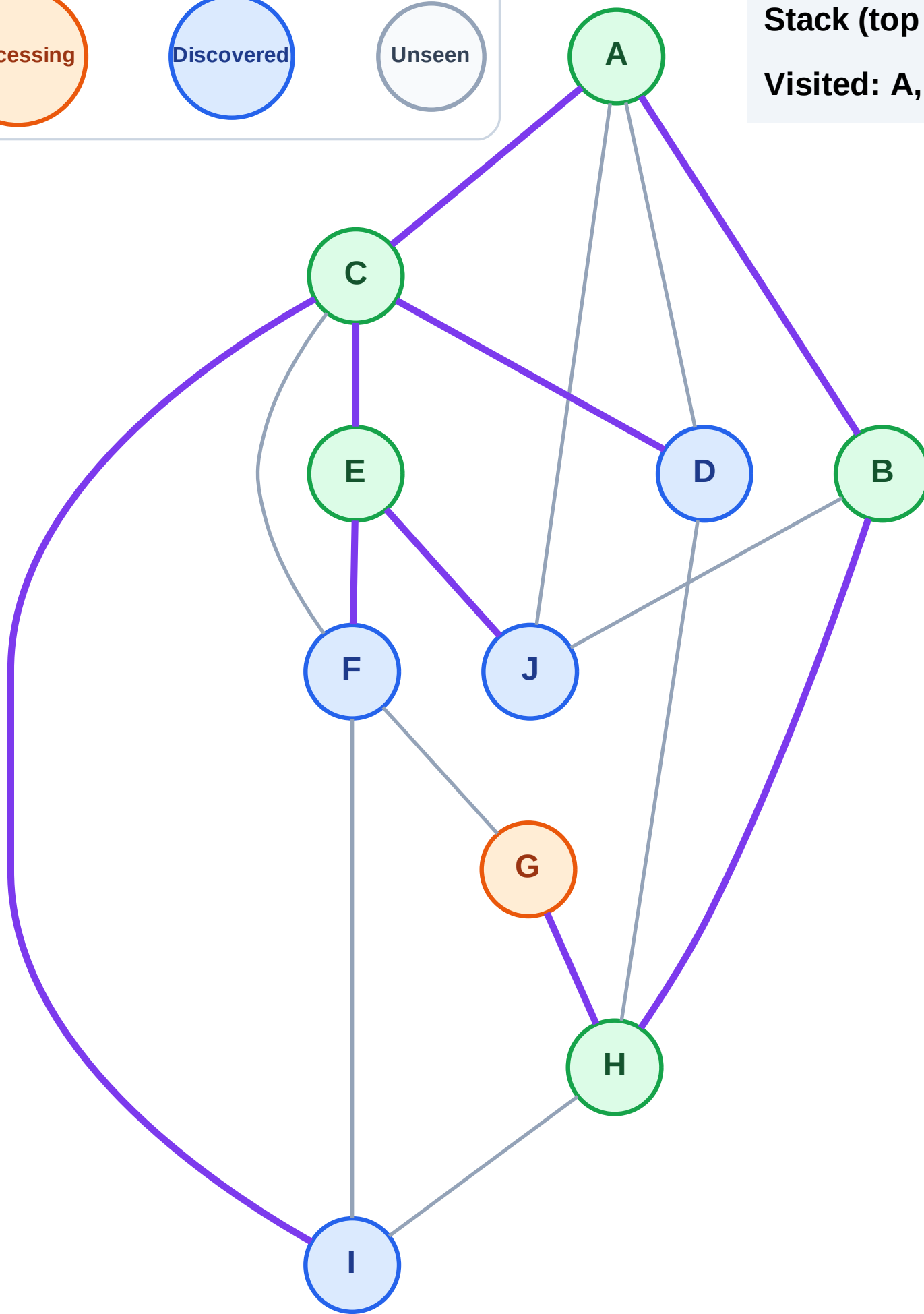
Visited: A, B, C, E, H



Step 12: Process node G



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DFS Traversal

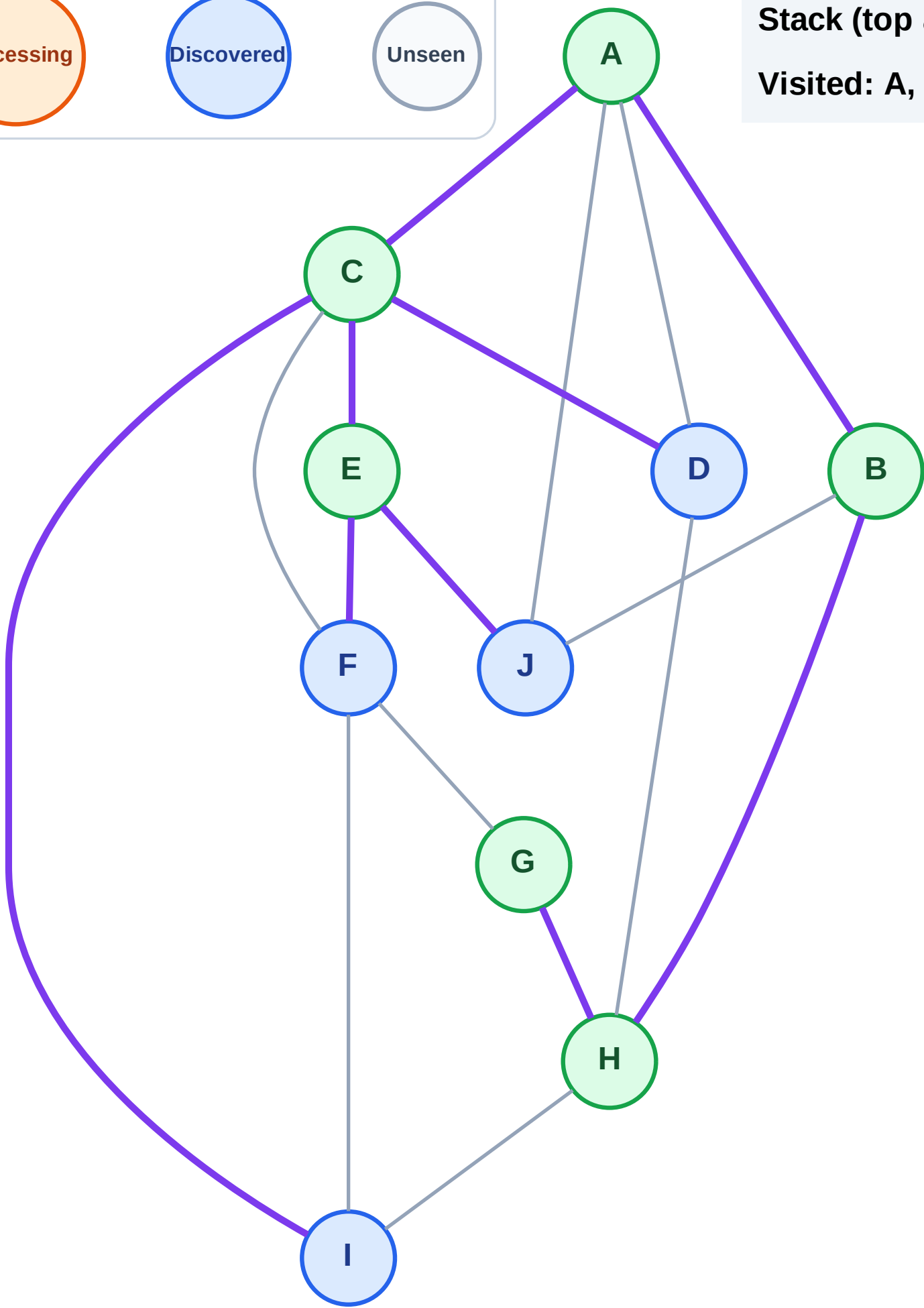
Step 13: No new neighbors from G

Legend



Stack (top at right): J | F | I | D

Visited: A, B, C, E, G, H



DFS Traversal

Step 14: Process node D

Legend

Complete

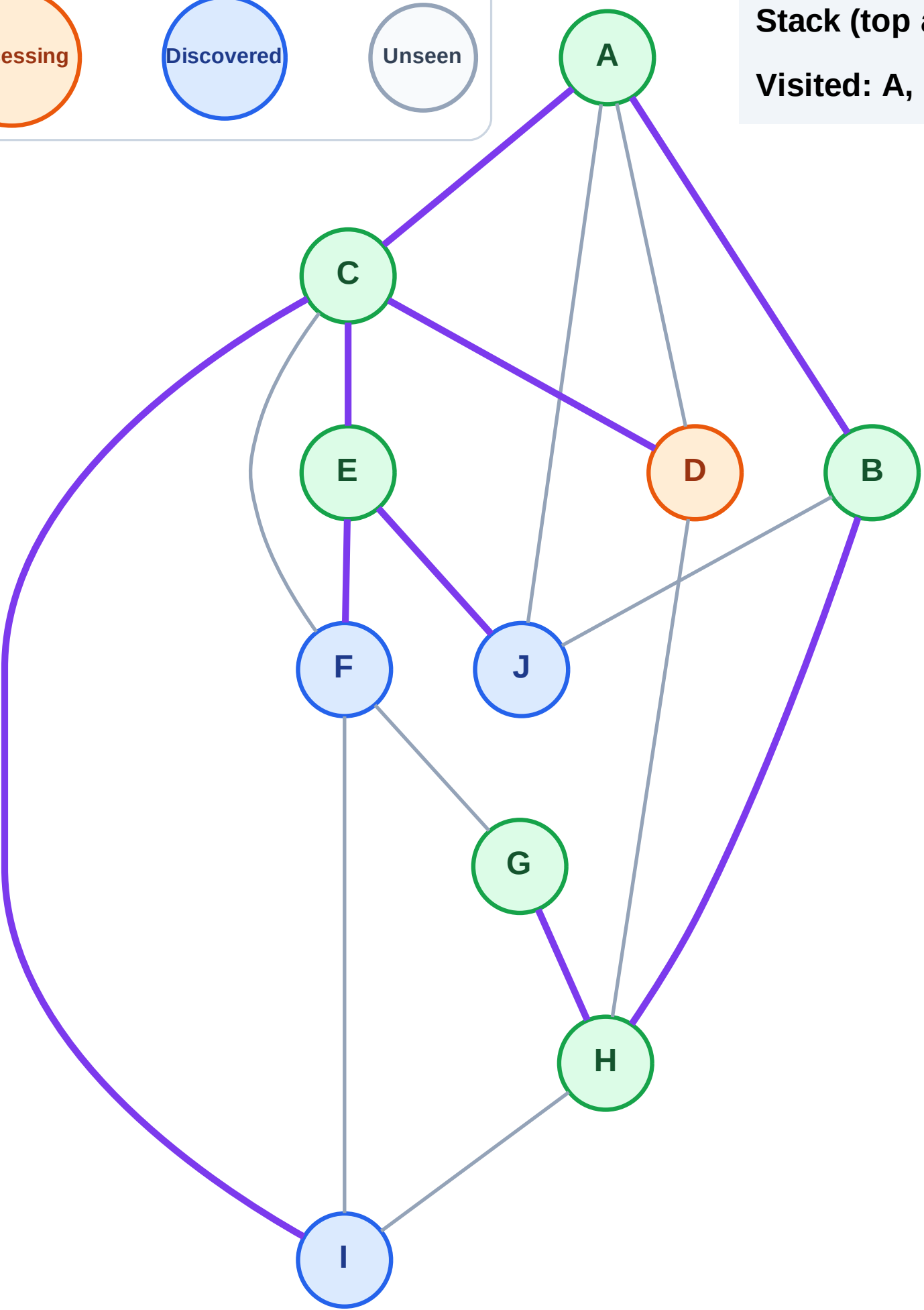
Processing

Discovered

Unseen

Stack (top at right): J | F | I

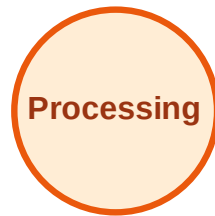
Visited: A, B, C, E, G, H



DFS Traversal

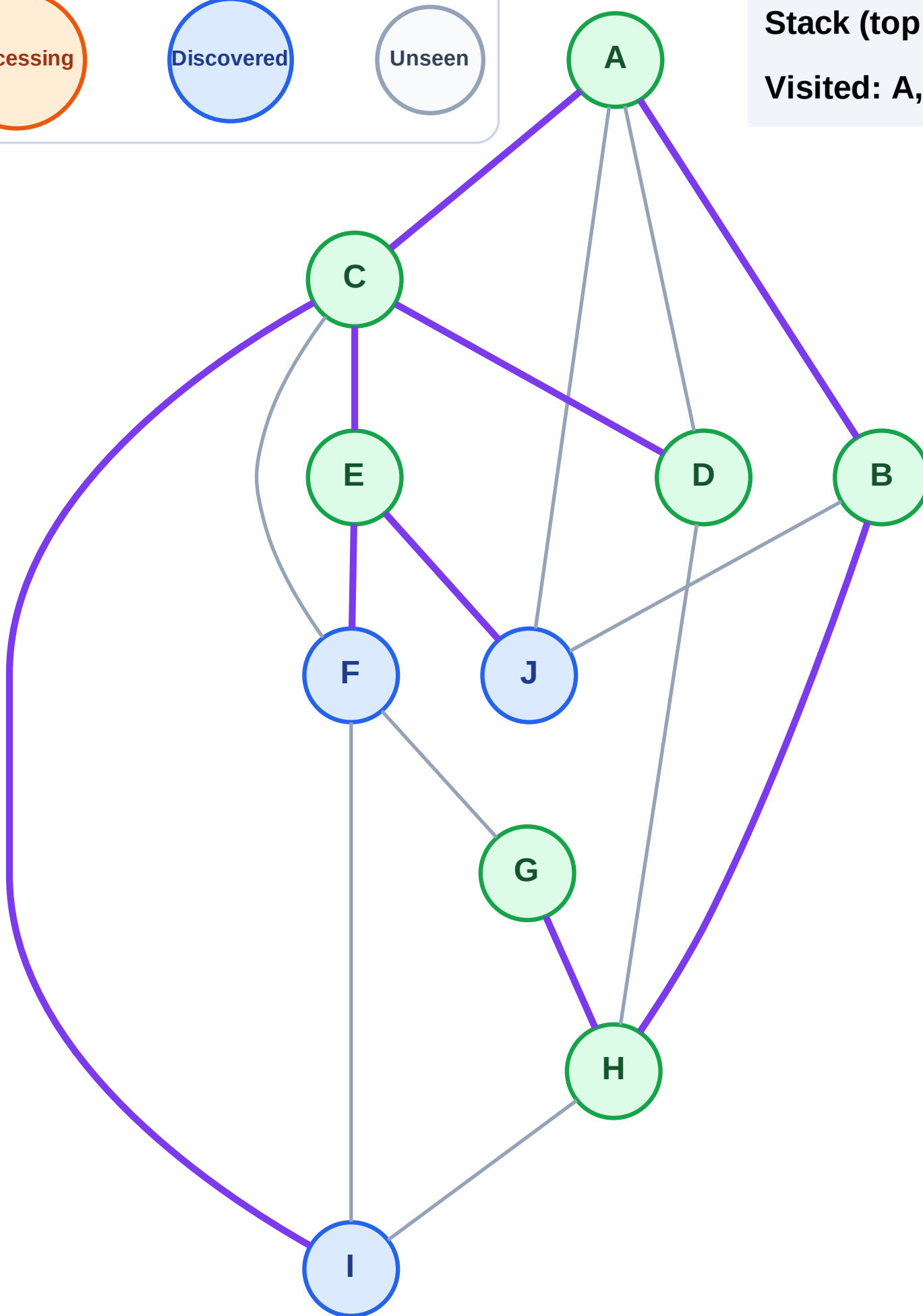
Step 15: No new neighbors from D

Legend



Stack (top at right): J | F | I

Visited: A, B, C, D, E, G, H



DFS Traversal

Step 16: Process node I

Legend

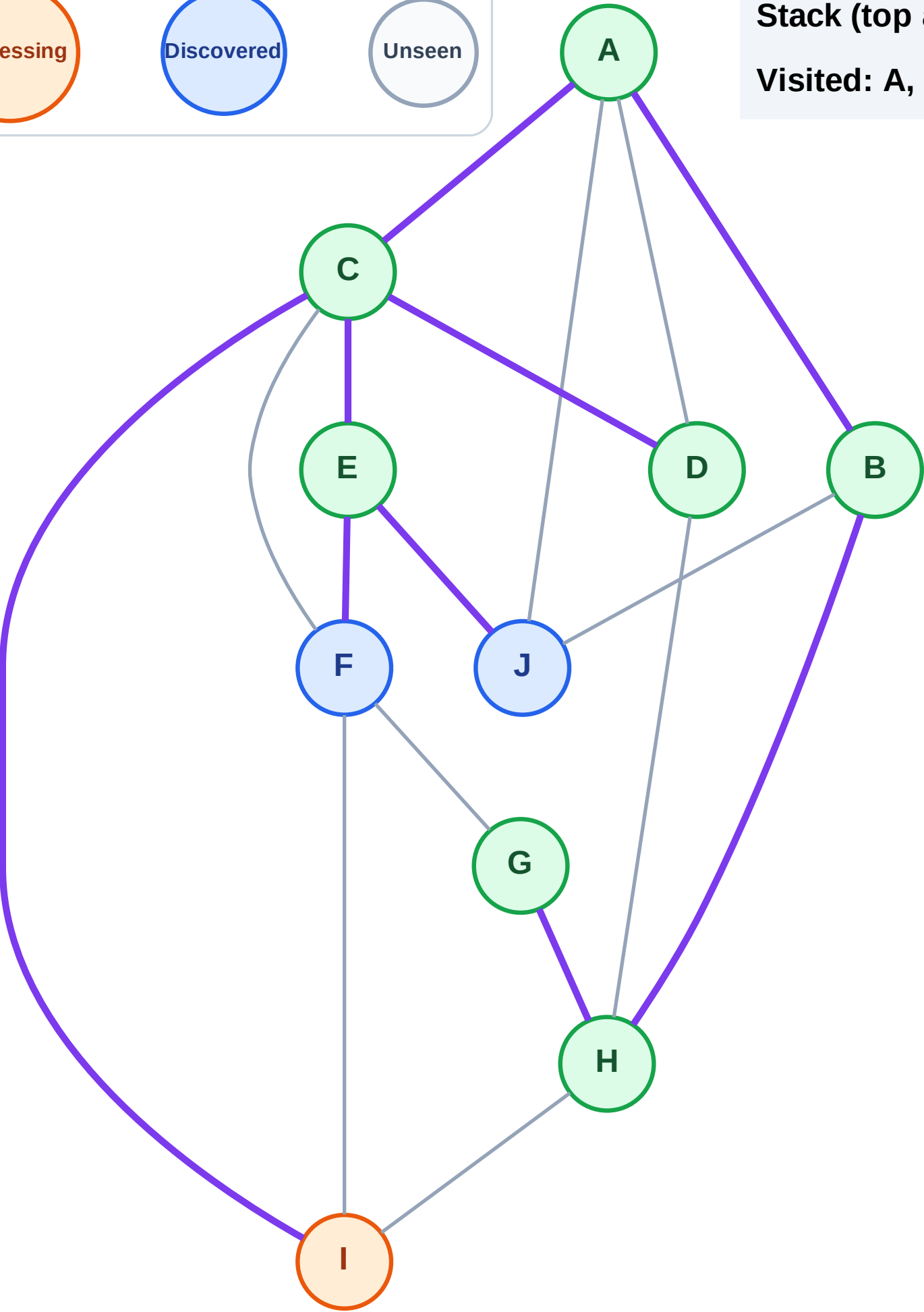
Complete

Processing

Discovered

Unseen

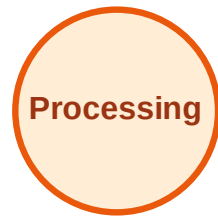
Stack (top at right): J | F
Visited: A, B, C, D, E, G, H



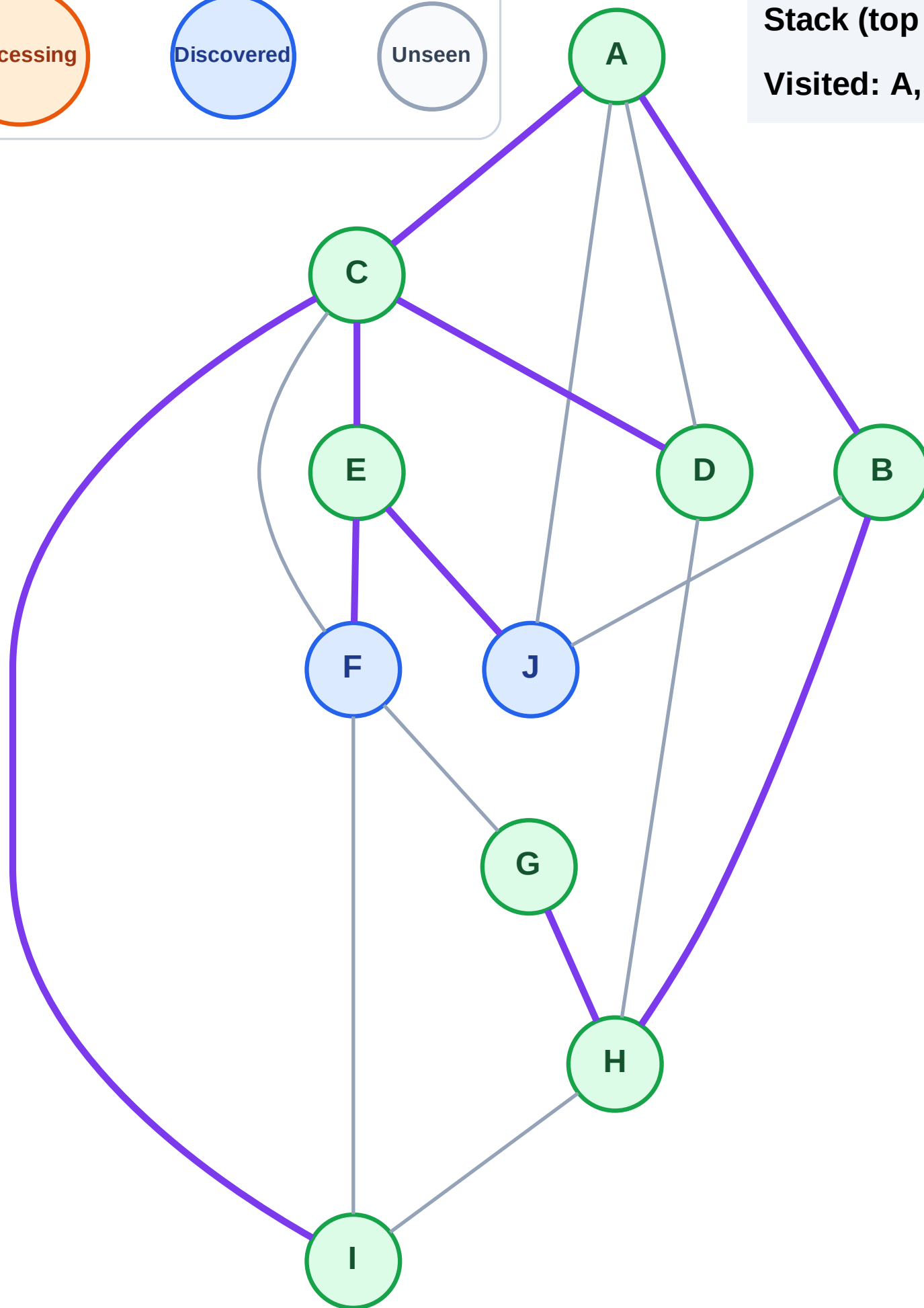
DFS Traversal

Step 17: No new neighbors from I

Legend



Stack (top at right): J | F
Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 18: Process node F

Legend

Complete

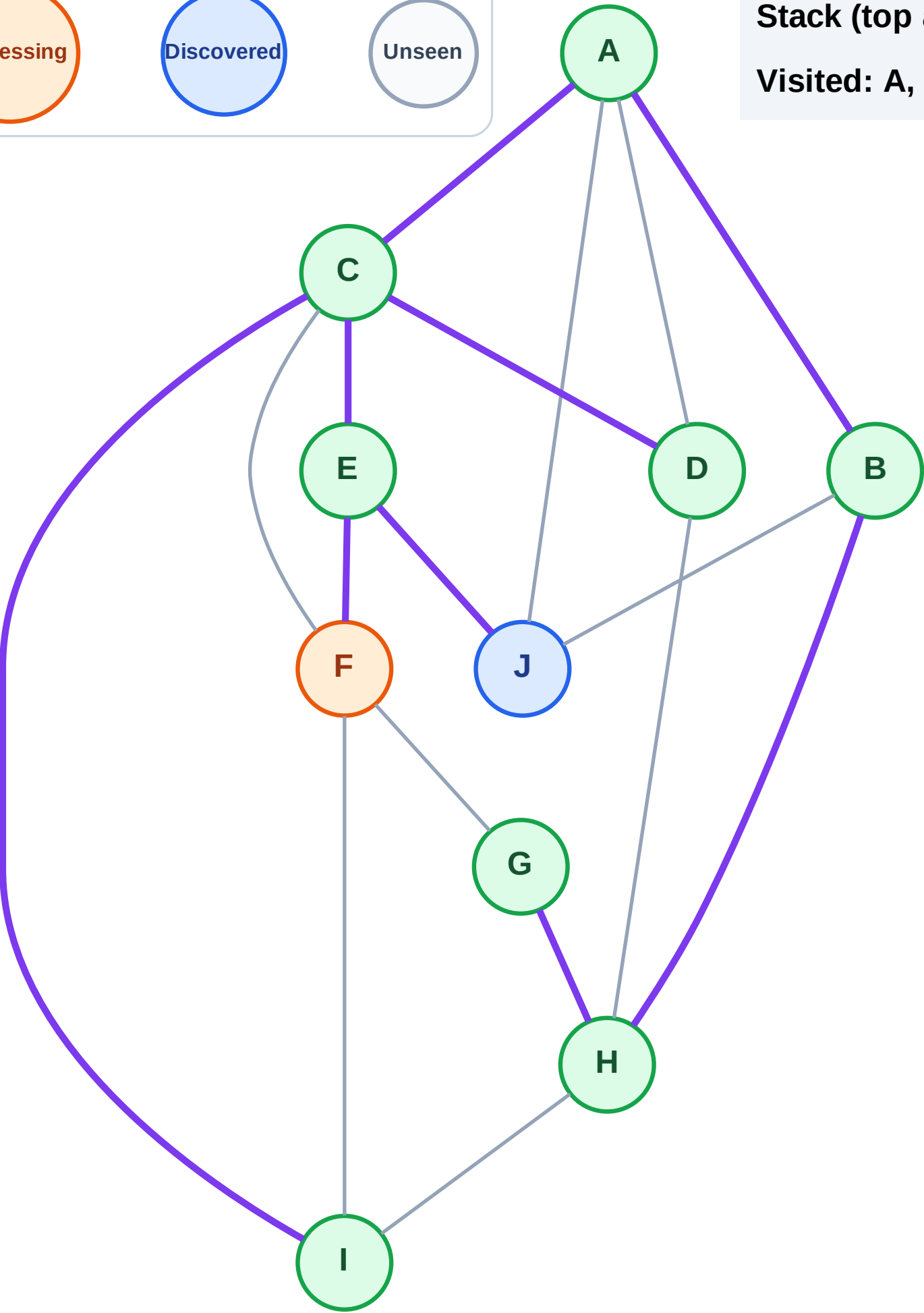
Processing

Discovered

Unseen

Stack (top at right): J

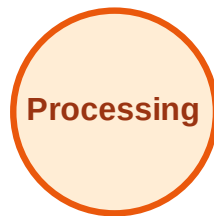
Visited: A, B, C, D, E, G, H, I



DFS Traversal

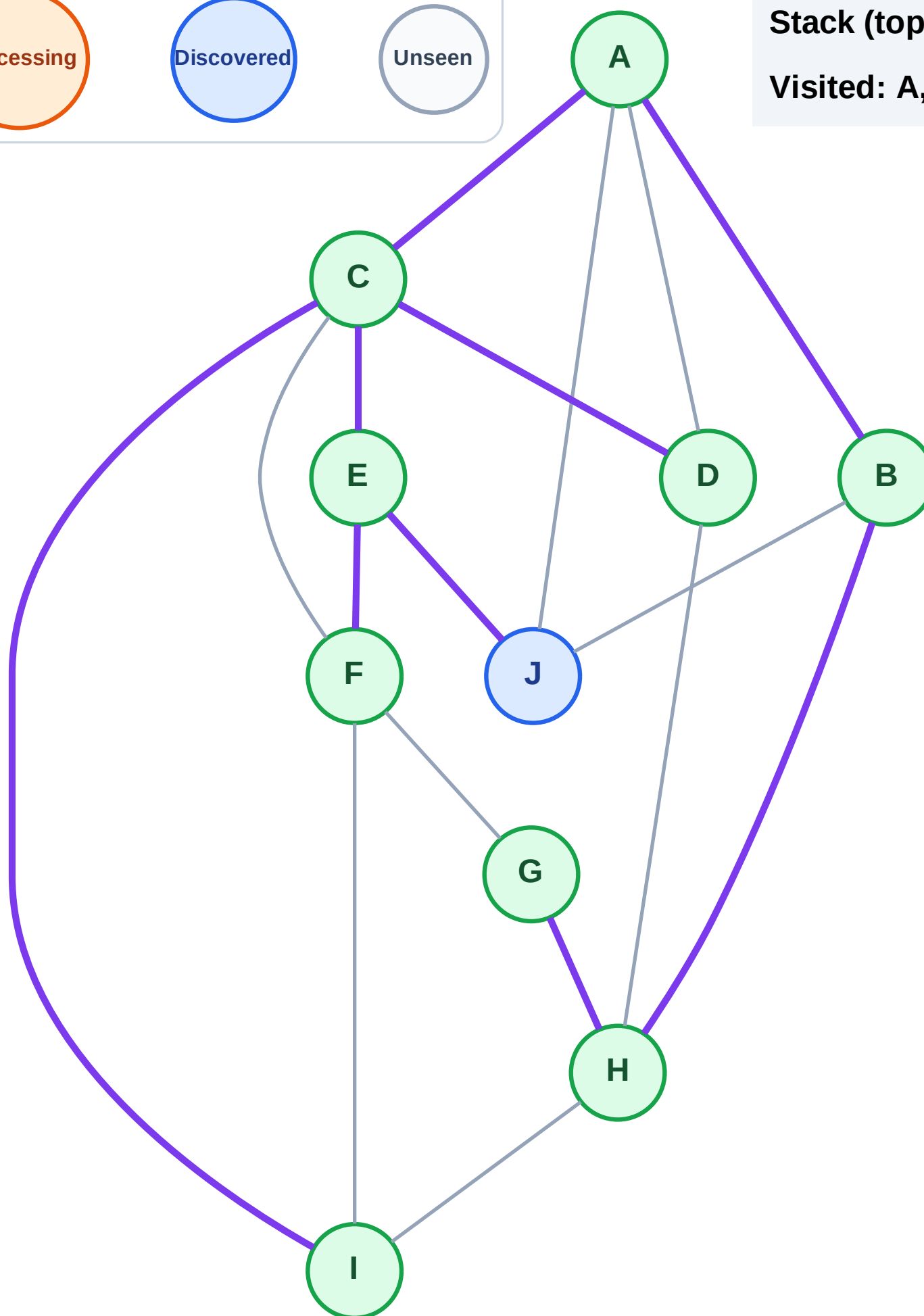
Step 19: No new neighbors from F

Legend



Stack (top at right): J

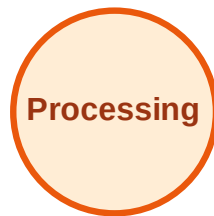
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

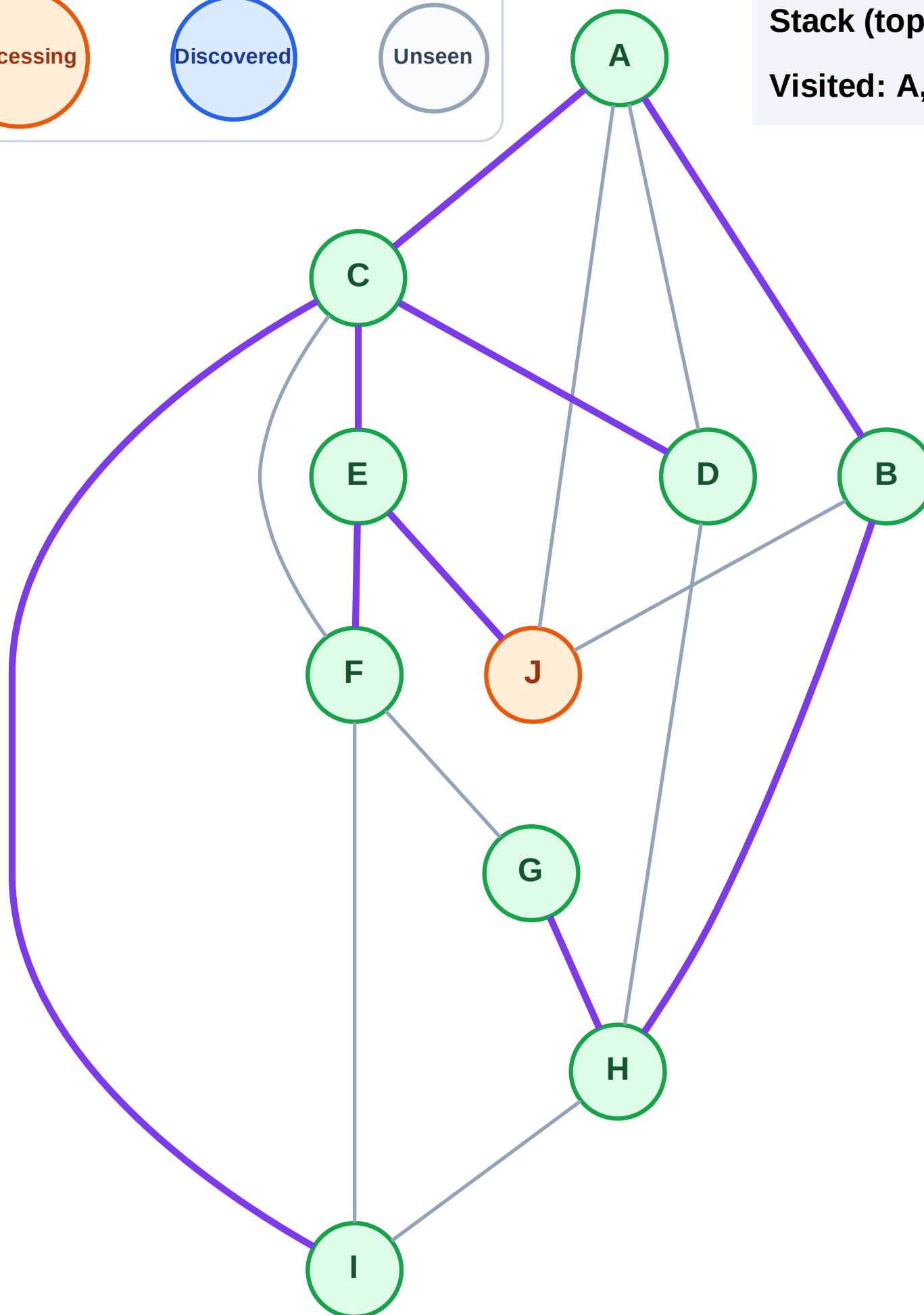
Step 20: Process node J

Legend



Stack (top at right): (empty)

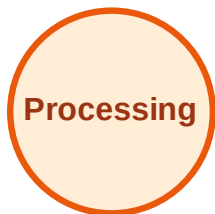
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

